

Coordinated Plaque Remodeling Modalities

A Falsifiable Synthesis from Material-Specific Plaque Remodeling to Sleep-Anchored, Low-Cost State-Setting

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IN BRIEF Problem. Existing plaque-regression literatures explain lipid lowering, efferocytosis, inflammation resolution, calcification and cap repair largely in parallel, but do not specify when improvement in one material-handling route is compatible with safe remodeling of the others. **Theory.** CPRMH treats plaque regression as a *coordination problem* across material-specific routes, governed by positive, negative and allied modalities plus a downstream safety/repair gate. **Novel prediction.** An intervention can improve one plaque axis yet fail to produce safe regression when another load-bearing axis is mismatched or a pre-specified negative modality remains active. CPRMH therefore uses a *pre-specification stack*: the stage-specific load-bearing set, admissible negative-modality blocker set, measurement model, and decision thresholds are fixed before outcome access. Coordination may appear as parallel effect modification or serial causal dependency; neither a single favorable biomarker, a post-hoc blocker, a post-hoc assay objection, nor a threshold chosen after the result counts as support. **Translation.** SAGDRC operationalizes recurrent state-setting using stable-feasible sleep anchoring, sleep-relative timing, embedded movement and low-cost local foods, explicitly including shift work, time poverty and access as exposure constraints. An adapted renal–metabolic branch keeps obesity, type 2 diabetes and CKD G3 inside the framework rather than treating them as exclusions. **Status.** The architecture is falsifiable and mechanistically grounded but is not a treatment recommendation; the complete chain remains to be tested in humans.

Abstract

Atherosclerotic plaque is a multi-material, multi-cellular structure containing retained apoB lipoproteins, intracellular and extracellular cholesterol, apoptotic and necrotic material, extracellular matrix, inflammatory cells and calcium-phosphate mineral in different phases. These components are often studied through separate literatures. This creates a theoretical gap: **what combination of material-handling processes would allow an established plaque to remodel while preserving or improving structural stability?**

This article develops the **Coordinated Plaque Remodeling Modality Hypothesis (CPRMH)** through a Theoretical Integrative Review and Perspective Synthesis. CPRMH distinguishes six positive remodeling modalities: suppression of new mineral formation; spatially confined mineral resorption; molecular cholesterol export; efferocytosis; active inflammation resolution; and structural stabilization/repair. It contrasts these with negative modalities such as diffuse acidification, protease spillover, foam-cell export failure, defective efferocytosis, mature-mineral persistence and excessive apoB inflow, while allied modalities alter the conditions under which positive remodeling can occur. The central proposition is that **safe regression is a coordination problem, not the maximization of any single clearance pathway**. Coordination is defined as prospectively specified causal dependency among stage-specific load-bearing axes and may be expressed through effect modification, serial mediation, perturbational blockade, or convergent combinations of these signatures. A complementary **Negative-Modality Lock** prospectively specifies which failure states are allowed to block the predicted transition, preventing N1–N9 from becoming a post-hoc rescue layer. A complementary **Measurement-and-Threshold Lock** pre-specifies how target engagement and blocker control are measured and how thresholds are adjudicated, preventing failed predictions from being rescued by retrospective claims that an assay was inadequate or a cutoff should have been moved.

The synthesis resolves several apparent contradictions. Proton-mediated mineral dissolution is chemically plausible, but diffuse plaque acidity can impair cholesterol efflux and amplify inflammation;

therefore the relevant variable is spatially confined, mineral-facing proton activity rather than systemic or diffuse acidification. Lipid removal is routed through molecular cholesterol export rather than bulk dissolution. Efferocytosis and pro-resolving signaling reduce secondary necrosis, but clearance is only favorable when structural repair keeps pace. Likewise, increasing calcium density can accompany plaque stabilization, so total calcium loss is not treated as a universal therapeutic endpoint.

A second proposition, the **Sleep-Anchored Grassroots Daily Remodeling Cycle (SAGDRC)**, asks whether recurrent low-cost behaviors can alter the probability of entering favorable modalities. Instead of fixed wall-clock times, SAGDRC anchors food withdrawal, fasting, caffeine withdrawal and activity to the individual's principal sleep–wake episode, while retaining endogenous circadian phase as a competing biological explanation. The revised framework separates sleep-anchor stability from anchor position and includes a renal–metabolic adapted branch for obesity, type 2 diabetes and CKD G3, in which prolonged fasting is not mandatory and diet, activity and monitoring are adjusted to renal and medication constraints. Embedded movement and ordinary low-cost foods are treated as first-line implementation resources rather than requiring protected exercise time, supplements or specialty foods.

Recent preclinical evidence that intermittent fasting can regress established Ldlr-deficient mouse plaque through a BMAL1–MERTK efferocytosis axis strengthens one state-setting branch of the framework but does not establish the full CPRMH chain or human efficacy. Accordingly, CPRMH specifies rival explanations, boundary conditions, seventeen falsification tests, a stage-specific pre-specification stack for load-bearing axes, negative-blocker sets, measurements and decision thresholds, and a staged empirical program. Safe-regression adjudication further requires prospectively justified harm margins for critical structural-safety endpoints rather than treating a non-significant safety test as evidence of no harm. Its intended contribution is a **testable coordination architecture** that connects plaque material routing, temporal physiology and health-equity constraints while pre-

serving strict separation between established evidence, cross-literature synthesis and new hypotheses.

KEYWORDS atherosclerosis; plaque regression; vascular calcification; macrophage; cholesterol efflux; efferocytosis; inflammation resolution; fasting; circadian biology; sleep regularity; chrononutrition; obesity; type 2 diabetes; chronic kidney disease; health equity; theoretical integrative review; perspective synthesis.

1. Introduction: From “Removing Plaque” to Coordinating Plaque Remodeling

Atherosclerosis is not a single-material obstruction. An advanced plaque can contain retained apoB-containing lipoproteins, extracellular and intracellular cholesterol, cholesterol crystals, apoptotic and necrotic cellular material, extracellular matrix, vascular smooth muscle cells (VSMCs), inflammatory cells, and calcium-phosphate mineral in multiple structural states [68, 70]. Consequently, the intuitive question—*how can plaque be cleaned?*—is biologically underspecified.

Three common simplifications create problems.

First, **plaque regression is often imagined as bulk material dissolution**. This is misleading for lipid. The physiologically plausible route for removing excess macrophage cholesterol is molecular export through transporters such as ABCA1 and ABCG1 to extracellular acceptors, followed by systemic lipoprotein transport, rather than the release of macroscopic lipid fragments into blood [12, 14, 70].

Second, **vascular calcium is frequently treated as one homogeneous substance**. In reality, calcium-phosphate deposition is a dynamic process involving nucleation, amorphous or poorly crystalline phases, crystal maturation, hydroxyapatite accumulation, microcalcification, macrocalcification, and variable mineral density. The biological significance of calcium also depends on location and morphology: microcalcification in mechanically vulnerable regions is not equivalent to dense macrocalcification within a more stable fibrocalcific plaque [27, 29, 30, 68].

Third, **acid-mediated demineralization is often conceptualized without spatial scale**. Hydroxyapatite dissolution is strongly pH-dependent, and osteoclasts use proton secretion to resorb mineral. Yet acidic extracellular plaque fluid can impair cholesterol efflux and amplify inflammation and proteolysis [8, 9]. The biologically relevant distinction may therefore be not whether acid is present, but **whether proton activity is confined to a sealed mineral-facing microcompartment or diffuses across the plaque environment**.

These observations suggest a different object of inquiry:

What combination of cellular and material-handling modalities would permit an established atherosclerotic plaque to remodel gradually while preserving or improving structural stability?

A second question follows:

Can recurrent everyday metabolic states—fasting, feeding, exercise, and habitual dietary exposures—alter the probability that these positive or negative remodeling modalities occur?

This article develops a conceptual response to these questions by integrating vascular mineral biology, macrophage resorption, cholesterol-efflux biology, efferocytosis, inflam-

mation resolution, plaque stabilization, and fasting/ketone physiology.

The proposed contribution is **not** the claim that fasting, weak acids, or a specific food “cleans arteries.” Instead, the article proposes that plaque remodeling should be analyzed as a **coordination problem across distinct modalities**.

2. The Analytical Problem: Mechanistic Knowledge Without a Shared Remodeling Architecture

The literature already provides strong mechanistic accounts of several components of atherosclerosis:

- VSMCs can undergo osteogenic transformation and promote vascular calcification.
- Calcium-phosphate crystal growth is actively inhibited by endogenous regulators such as pyrophosphate.
- Macrophages can both promote calcification and participate in ectopic demineralization.
- Macrophages export cholesterol through ABCA1/ABCG1-dependent pathways.
- Advanced plaques show defective efferocytosis and accumulation of necrotic material.
- Specialized pro-resolving mediators can improve efferocytosis and plaque stability in experimental models.
- Intensive lipid lowering can reduce non-calcified plaque burden while increasing or densifying calcified components.
- BHB can inhibit vascular calcification in several preclinical models.
- Fasting and time-restricted feeding can improve or worsen atherosclerosis depending on metabolic and genetic context.

The problem is therefore **not absence of mechanisms**.

The problem is one of **analytical organization**.

Most literatures ask questions such as:

- Does BHB inhibit calcification?
- Does macrophage CA2 dissolve ectopic mineral?
- Does ABCA1 promote cholesterol efflux?
- Does MerTK-mediated efferocytosis reduce necrotic core?
- Do resolvins improve plaque stability?
- Does time-restricted feeding reduce atherosclerosis?
- Does higher calcium density indicate stabilization?

Each question is important, but the literatures rarely ask what must happen **simultaneously or sequentially** if a plaque is to lose hazardous material without becoming mechanically unstable.

The missing explanatory object is therefore proposed as:

Coordinated plaque remodeling across mineral, lipid, cellular-debris, inflammatory, and structural compartments.

3. Conceptual Gap

3.1 Material-specific clearance gap

Plaque components are often aggregated under broad endpoints such as “plaque burden” or “calcification,” but different materials require different removal routes (Table 1).

A single “cleaning” mechanism cannot safely perform all of these tasks.

3.2 Spatial-acidification gap

Acidity is generally discussed as either pro-atherogenic or mineral-dissolving. Both can be true at different spatial scales.

Table 1. Material-specific plaque handling routes

Plaque component	Plausible handling route
Calcium-phosphate mineral	ion release after mineral dissolution/resorption
Intracellular cholesterol	lipophagy / lysosomal hydrolysis → ABCA1/ABCG1-mediated efflux
Extracellular aggregated lipoprotein	digestive exophagy, uptake, and subsequent cellular cholesterol handling
Apoptotic cells	efferocytosis
Necrotic debris	phagocytic/efferocytic clearance plus resolution
Extracellular matrix	regulated remodeling, not indiscriminate degradation

- **Diffuse extracellular acidity** can impair ABCA1-mediated cholesterol efflux, promote LDL modification, and increase inflammatory/proteolytic activity [8, 9].
- **Localized acidic compartments** are used by osteoclasts for mineral resorption and by macrophages in lysosomal synapses for extracellular digestion [10, 11, 12].

The crucial variable may therefore be **spatial confinement**, not simply pH.

3.3 Clearance–repair coordination gap

Material removal alone is insufficient. If mineral or matrix is removed faster than VSMCs can maintain collagen and the fibrous cap, plaque vulnerability may increase. Efferocytosis, pro-resolving signaling, and structural repair must therefore be incorporated as co-requisites rather than downstream details.

3.4 Stage-dependence gap

Early macrophages, advanced foam cells, calcified plaques, diabetes-associated lesions, and CKD-associated mineralization are biologically different states. A pathway available in one stage may be absent in another. For example, GPR109A signaling is down-regulated in human macrophage-derived foam cells, limiting extrapolation from receptor-mediated anti-inflammatory or efflux mechanisms in less lipid-loaded cells.

3.5 Everyday-state translation gap

Fasting, meal timing, exercise, and diet are often treated as interventions in their own right. Yet these practices may be better conceptualized as **state-setting exposures** that alter ketones, insulin, glucose, lipoprotein flux, inflammatory signaling, substrate availability, and circadian physiology. Their relevance to plaque remodeling depends on whether they actually shift the underlying cellular modalities.

3.6 Feasibility, time-cost, and health-equity gap

A lifestyle exposure can be biologically plausible yet socially unusable. Conventional cardiovascular advice often assumes discretionary time, stable work hours, access to safe exercise space, separate meal-preparation time, and the ability to organize sleep around a daytime work schedule. These assumptions exclude or burden shift workers, caregivers, informal workers, low-income households, and people whose work already consumes most of their physical and temporal capacity.

The relevant translational object is therefore not merely *recommended behavior* but **feasible health-maintenance possibility**: what can be incorporated into a person's existing life without requiring substantial additional money, travel, equipment, or protected leisure time. This distinction parallels the difference between nominal opportunity and practically usable opportunity developed in social analyses of sporting possibility: time, work, care, transport, and institutional organization shape

what a person can actually do, not merely what is theoretically available (Lahtee, 2026).

This article therefore treats **time itself as a health resource**. The grassroots extension of CPRMH prioritizes behaviors that can be embedded into ordinary activity and schedules biological transitions relative to sleep and wake rather than to socially standardized clock times.

4. Nearest Neighbors and the Discriminating Novelty of CPRMH

The novelty claim is deliberately narrower than saying that plaque regression, efferocytosis, calcification, meal timing or fasting are individually new. Contemporary scientific statements already describe regression as involving lower lipid burden, reduced inflammation, improved efferocytosis, cap thickening and changes in calcific phenotype [68, 69]. Reviews also connect macrophage phenotype, efferocytosis and vascular calcification [6, 7, 70], while chrononutrition literature increasingly links feeding time to cardiovascular and circadian physiology [67].

The proposed advance is the **analytical object**: CPRMH asks whether these processes form a constrained, material-specific remodeling system in which the benefit of one route depends on the state of the others. Table 2 shows the nearest-neighbor challenge.

Discriminating novelty. CPRMH earns theoretical complexity only if a multi-modality coordination term explains safe plaque remodeling beyond what can be explained by apoB reduction, weight loss, glycemic improvement, circadian alignment, isolated efferocytosis or formation suppression alone.

5. Core Concept: The Coordinated Plaque Remodeling Modality Hypothesis

The **Coordinated Plaque Remodeling Modality Hypothesis (CPRMH)** proposes that:

Safe regression or stabilization of an atherosclerotic plaque depends less on maximizing any single clearance mechanism than on coordinating material-specific clearance, suppression of new deposition, inflammation resolution, and structural repair while preventing diffuse acidification, proteolytic spillover, and cellular overload.

Definition of modality. A *modality* is a context-dependent operational configuration of the remodeling system. It is not a molecule, intervention or competing model. Positive, negative and allied modalities may coexist in different cells, plaque regions and times.

Proposition C0 — Coordination Necessity. If CPRMH is correct, improving one material-handling axis will not reliably produce safe regression when another *stage-specific load-bearing axis* remains

Table 2. Nearest-neighbor literatures and the narrower novelty claim of CPRMH

Neighbor	What it already explains	What CPRMH adds as a testable object
Plaque-regression statements [68, 69]	lipid reduction, inflammation, cap change, imaging-defined regression	material-specific exit routes plus an explicit coordination/safety gate; regression need not imply parallel calcium loss
Macrophage/efferocytosis literature [15, 16, 70]	apoptotic-cell clearance, necrotic-core control, inflammatory resolution	asks whether efferocytosis remains beneficial when lipid export, mineral handling and matrix repair are mismatched
Vascular-calcification literature [4, 6, 7]	mineral formation, macrophage–VSMC interactions, micro/macrocacification	separates formation suppression from mature-mineral removal and spatially confined resorption from diffuse acidity
Chrononutrition/fasting literature [24, 66, 67]	feeding-time, circadian, metabolic and emerging plaque effects	treats repeated metabolic exposures as probabilistic state setters for a multi-modality plaque system rather than as direct plaque-cleaning interventions
Health-behavior guidance [42, 43]	recommended diet and physical-activity patterns	makes time, cost, shift work and feasibility part of exposure validity and tests embedded low-cost implementation

Note. The table does not claim that the component mechanisms are novel. The novelty claim concerns their reorganization into a falsifiable coordination architecture and the predictions generated by that reorganization.

limiting. The required dependency need not always appear as a conventional statistical interaction. It may take the form of **parallel conditional dependency** (effect modification) or **serial causal dependency** in which one modality must transmit or permit the effect of another.

Proposition C1 — Recurrent Permissive Windows. Repeated everyday state-setting exposures may influence plaque trajectory by changing the probability and duration of favorable modality configurations. Their explanatory value should survive adjustment for simpler mediators such as energy intake, weight, apoB, glycemia and circadian phase.

Proposition C2 — Stage-Specific Architecture and Causal-Signature Lock. For every empirical test, the set of modalities asserted to be load-bearing for the studied plaque stage, together with the expected dependency among them, must be specified **before outcome access**. A failed interaction cannot be rescued by relabeling the relation as mediation after analysis, and a failed load-bearing axis cannot be deleted from the same dataset. Any revised architecture constitutes a new theory version that requires prospective testing in independent data.

Proposition C3 — Stage-Specific Negative-Modality Lock and Exhaustion. For every empirical test, the negative modalities permitted to block the predicted transition must also be specified **before outcome access** as a stage-specific blocker set B_s . Each blocker requires an operational measure, expected direction, threshold or decision rule, temporal position, and declared role (blocker, modifier, mediator, or safety failure). If the locked positive architecture shows target engagement, the locked causal signatures are observed, and every relevant member of B_s is absent, controlled, or below its blocking threshold, failure of the pre-specified remodeling outcome counts as falsification of that stage-specific architecture rather than evidence for an unspecified new negative modality.

Proposition C4 — Measurement, Threshold, and Causal-Precedence Lock. For every empirical test, the measurements used to establish target engagement, blocker presence, causal intermediates, and structural safety must be specified **before outcome access**. The measurement model M_s must declare the assay or instrument, construct being measured, validity evidence available for the population / model, precision or repeatability criterion, spatial and temporal resolution, quality-control failure rule, and the minimum adequacy required for causal adjudication. The threshold set T_s must declare target-engagement, blocking, and harm thresholds together with a biologically defensible plausible range for sensitivity analysis. A blocker cannot be defined by the same downstream outcome that it is invoked to explain; its operational signal must precede or be causally upstream of that

outcome. Once M_s and T_s pass their pre-specified adequacy rules, they cannot be redefined after the result merely to rescue the current architecture.

Proposition C5 — Minimum Core Identity and Core-Collapse Criterion. The irreducible CPRMH core for established / advanced plaque is the claim that **safe remodeling requires causal dependence among at least two biologically separable load-bearing functions and that this dependency adds predictive or causal information beyond a functionally sufficient single-axis model**. The framework may contain stage-specific branches, but it cannot retain the CPRMH label in a declared core disease domain if orthogonal prospective tests repeatedly show either single-axis sufficiency or exhaustion failure under adequate measurement, robust thresholds, and pre-specified safety adjudication. In that circumstance the core coordination claim for that domain is falsified rather than merely “weakened.”

The pre-specification stack. For plaque stage s , CPRMH distinguishes four locked objects: L_s (load-bearing functions and causal edges), B_s (admissible negative-modality blockers), M_s (measurement model and adequacy criteria), and T_s (target-engagement, blocker, decision and harm thresholds). Together these specify what must work, what may block it, how those states will be observed, and what magnitude counts as biologically meaningful. A result cannot be rescued in the same dataset by changing any member of this stack after outcome access.

Two admissible causal signatures. CPRMH recognizes two primary forms of coordination. **Parallel conditional dependency** predicts that the marginal effect of axis A is attenuated when axis B is impaired and amplified when B is permissive. **Serial dependency** predicts that A changes B (or requires B to remain functional), and that perturbing or disabling B attenuates the downstream effect of A. Statistical mediation can help quantify the serial path, but mediation alone is not treated as proof of causal necessity without appropriate identification assumptions and, where feasible, perturbational or orthogonal evidence [77, 78].

The stage-specific lock. Let L_s denote the stage-specific load-bearing set, B_s the stage-specific negative-modality blocker set, M_s the measurement model, and T_s the decision-threshold set. All four are locked before outcome access. The purpose of B_s is not to require measurement of all N1–N9 in every study; only biologically relevant blockers for the declared stage and causal contrast are included. A critical blocker or load-bearing construct that does not meet its pre-specified M_s adequacy criterion makes the strongest causal adjudication **measurement-indeterminate**; it does not become a free post-

Table 3. Prospectively admissible causal signatures of coordination

Signature	Pre-specified prediction	What would weaken the claimed dependency
Parallel conditional dependency	the marginal benefit of axis A is attenuated when axis B is impaired and amplified when B is permissive, beyond the pre-specified additive model	repeated failure of the predicted direction in adequately powered data; switching scale, modifier, or direction after outcome access is not permitted
Serial causal dependency	A produces part of its effect through B, or requires B to remain functional; blocking or disabling B attenuates A's downstream plaque effect	the total effect of A persists despite effective perturbation of B, or prospectively specified mediation / perturbation analyses show no material dependency
Convergent dependency	effect modification and serial/perturbational evidence point to the same load-bearing relation	orthogonal tests repeatedly disagree in a way better explained by a simpler single-axis model

Note. The signature family is selected before outcome analysis. A failed interaction may motivate a future serial-dependency hypothesis, but it cannot be retrospectively counted as support in the same dataset. Causal mediation and interaction are distinct but can coexist; their decomposition has been formalized in counterfactual causal-inference frameworks [77, 78].

hoc explanation. Likewise, a conclusion that changes across plausible values in the pre-specified T_s sensitivity range is **threshold-fragile**, not strong confirmation.

Pre-specified negative modality = prediction. Post-hoc negative modality = rescue. CPRMH permits the first and prohibits the second for adjudicating the same dataset.

The framework distinguishes three classes.

Positive modalities. States or mechanisms that favor safe re-modeling:

1. suppression of new mineral formation;
2. spatially confined mineral resorption;
3. molecular cholesterol export;
4. effective efferocytosis;
5. inflammation resolution;
6. fibrous-cap preservation or repair;
7. where appropriate, transformation toward a denser, more stable calcific phenotype.

Negative modalities. States or mechanisms that block clearance or increase structural risk:

1. diffuse extracellular acidification;
2. incomplete osteoclast-like differentiation;
3. protease spillover into the fibrous cap;
4. foam-cell failure of cholesterol-export machinery;
5. defective efferocytosis;
6. mature-mineral persistence;
7. excessive apoB influx or repeated hyperlipidemic exposure;
8. disease-state mismatch;
9. state-setting exposures that worsen rather than improve downstream clearance.

Allied modalities. Mechanisms that are not themselves clearance but improve the environment in which positive modalities can operate:

1. apoB/LDL reduction;
2. pyrophosphate and MGP-related mineral-growth inhibition;
3. autophagy/lipophagy;
4. specialized pro-resolving mediator pathways;
5. restoration of hepatic lipoprotein clearance;
6. fasting or time-restricted eating under permissive metabolic conditions;
7. exercise and other metabolic-state transitions;

8. dietary patterns that reduce pro-calcific or pro-inflammatory pressure.

6. Positive Modalities

6.1 Positive Modality 1: Suppression of New Mineral Formation

Vascular calcification is not only passive precipitation. VSMC osteogenic transition, apoptosis, extracellular vesicles, inflammatory signals, phosphate load, and loss of endogenous inhibitors all contribute to mineral formation. Hydroxyapatite deposition therefore reflects both physicochemical permissiveness and active cell-regulated processes.

This distinction matters because **stopping new mineral formation is biologically easier than removing mature mineral**.

BHB has emerged as a candidate suppressor of vascular calcification in preclinical systems. Three partially independent mechanistic lines are notable:

1. **Autophagy enhancement.** BHB reduced high-phosphate-induced arterial and VSMC calcification, and inhibition of autophagy attenuated this protection.
2. **HDAC9-NF- κ B modulation.** BHB reduced VSMC and arterial calcification in CKD-related models while down-regulating HDAC9 and NF- κ B signaling.
3. **AMPK/Nrf2 and ER-stress modulation.** In ApoE-deficient mice and VSMCs, BHB reduced calcification and ER-stress-mediated apoptosis.

These results support a **formation-suppression modality**. They do not demonstrate mature mineral removal.

Ketone-independent pharmacologic triangulation. The formation-suppression arm is not dependent on BHB alone. Dapagliflozin has been reported to attenuate VSMC osteogenic transdifferentiation and vascular calcification through an SGLT2-SIRT1 pathway with SIRT1-mediated deacetylation and degradation of HIF-1 α [47]. Canagliflozin has independently reduced vascular calcification in CKD-rat, vitamin-D-overload mouse, arterial-ring, and VSMC models through suppression of the NLRP3-caspase-1-IL-1 β axis, with NLRP3 inhibition and knockdown providing mechanistic controls [48]. These findings are important to CPRMH because they show that a *formation-suppression modality* can be entered through more than one upstream state-setting route. They also prevent the theory from becoming a ketosis-only explanation.

Table 4. Prospective negative-modality lock: directional predictions and defeat conditions

Modality	Declared blocker role	Prospective directional prediction	What would weaken or defeat the negative claim
N1	diffuse-acidity / safety blocker	diffuse extracellular acidification impairs efflux or increases inflammatory / proteolytic injury relative to confined mineral-facing acidity	replicated diffuse acidification with preserved or improved efflux and structural safety in the relevant system
N2	mineral-resorption blocker	incomplete osteoclast-like machinery limits removal of established mature mineral even when formation falls	substantial mature-mineral removal despite prospectively verified incomplete resorptive machinery and no alternative resorptive route
N3	matrix-safety blocker	increased non-confined protease activity increases cap / matrix injury when resorptive activity rises	high target-engaged protease activity repeatedly occurs without deterioration beyond the locked structural harm margin
N4	lipid-export blocker	advanced foam-cell transformation attenuates the response of cholesterol-export machinery to the candidate state setter	advanced foam cells repeatedly recover strong export capacity under the locked exposure despite the predicted receptor / transporter limitation
N5	debris-clearance blocker	debris generation that exceeds efferocytic capacity increases secondary necrosis / necrotic burden	debris generation exceeds the locked efferocytic capacity yet necrotic burden falls through a reproducible alternative clearance route
N6	mineral-state blocker	mature apatite persists when only formation pressure is reduced	established mature mineral regresses materially without target engagement of any pre-specified resorptive mechanism
N7	context modifier / state-setting blocker	the same fasting exposure changes direction across pre-specified genetic / metabolic / hepatic-clearance contexts	a locked eligible context yields null or opposite effects and cannot be rescued by a new post-hoc subgroup
N8	inflow-conflict blocker	a state setter that raises apoB can offset or reverse gains in other modalities when pathological inflow becomes dominant	sustained large apoB increase repeatedly coexists with safe regression under adequate control of alternative explanations, requiring revision of the hydraulic principle
N9	temporal blocker	unstable sleep anchoring reduces the benefit of the relative-time architecture beyond total sleep / energy effects	anchor stability adds no reproducible signal beyond circadian phase, sleep duration and energy intake

Note. A negative modality is a prediction, not a universal excuse. The relevant subset, operationalization and expected direction are frozen before outcome access. A newly discovered blocker can motivate the next theory version but cannot adjudicate the dataset in which it was discovered.

Endogenous mineral inhibitors provide an independent allied pathway. Pyrophosphate is a potent inhibitor of hydroxyapatite formation and growth; genetic disruption of extracellular PPI metabolism can cause severe arterial calcification in humans. The significance of this pathway is conceptually important: the vascular wall possesses active anti-mineralization systems that can shift mineral balance without requiring dissolution of existing mineral.

Proposition P1. A metabolic state may reduce vascular calcification progression primarily by lowering mineral initiation and propagation rather than by increasing resorption of mature mineral.

6.2 Positive Modality 2: Spatially Confined Mineral Resorption

The chemical premise is straightforward: calcium-phosphate mineral becomes more soluble as proton activity increases. Bone resorption demonstrates a biological implementation of this chemistry. Osteoclasts attach to mineralized surfaces, create a sealing zone, acidify the resorption lacuna through V-ATPase-dependent proton transport, and dissolve hydroxyapatite while the surrounding extracellular environment remains near physiological pH.

Evidence suggests macrophages possess parts of a related machinery. Barinda and colleagues reported high macrophage expression of carbonic anhydrase 2 (CA2) and demonstrated macrophage-mediated demineralization of ectopic calcification in vitro [5]. Inflammatory polarization and angiotensin II

reduced CA2 expression and decalcification capacity, whereas CA2 overexpression increased demineralization.

This is an important proof of principle:

The body does not need systemic acidosis to perform acid-mediated mineral resorption.

However, this modality is incomplete in atherosclerotic plaque. Human plaque macrophages may express CA2 yet show weak cathepsin K and defective osteoclast-like differentiation [6, 7]. Other experimental work suggests mature apatite can remain persistent even after the original pro-calcific metabolic environment is corrected.

The hypothesis therefore requires a strict spatial safety condition:

Useful acidification must be mineral-facing, highly localized, transient, and physically or functionally separated from the broader plaque environment.

Proposition P2. Spatial confinement of proton activity is more important than systemic or diffuse plaque acidity for any potentially beneficial acid-mediated vascular demineralization.

6.3 Positive Modality 3: Molecular Cholesterol Export

Safe plaque remodeling requires lipid to leave the lesion without bulk lipid fragments entering the bloodstream. Macrophage cholesterol-efflux pathways provide this mechanism.

Excess intracellular cholesteryl ester can be hydrolyzed to free cholesterol. ABCA1 transfers cholesterol and phospho-

lipid to lipid-poor apoA-I, forming nascent HDL-like particles, while ABCG1 and related mechanisms support efflux to more mature HDL particles. These processes constitute the first steps of reverse cholesterol transport.

This route is conceptually different from “dissolving fat.” The material exits through **molecular transfer**, not mechanical detachment.

Autophagy and lysosomal lipid metabolism may assist this route. Lipophagy can mobilize lipid droplets, lysosomal acid lipase can hydrolyze stored lipid, and the released cholesterol can become available for ABCA1-dependent export.

However, advanced foam cells can lose this capacity. Free-cholesterol loading accelerates ABCA1 degradation, and human plaque macrophage phenotypes can display poor cholesterol handling despite high phagocytic activity.

Proposition P3. Any plaque-remodeling modality that increases lipid digestion without preserving downstream cholesterol export risks converting extracellular lipid into intracellular foam-cell burden rather than achieving net plaque clearance.

6.4 Positive Modality 4: Efferocytosis and Cellular-Debris Clearance

Advanced plaques contain dying macrophages and other cells. If apoptotic cells are not removed rapidly, they undergo secondary necrosis and contribute to necrotic-core expansion. Efferocytosis therefore represents a separate material-clearance task.

MerTK is a key efferocytosis receptor. Experimental prevention of MerTK cleavage improves efferocytosis, reduces necrotic-core size, increases fibrous-cap thickness, and shifts lipid-mediator balance toward resolution. Human carotid plaque evidence associates MerTK cleavage with necrosis and ischemic symptoms.

This provides a crucial safety principle:

Material mobilization without cellular-debris clearance can worsen plaque biology even if one component of plaque burden declines.

Proposition P4. Net plaque remodeling requires clearance capacity to rise at least as fast as the generation or exposure of apoptotic, necrotic, or lipid-derived material.

6.5 Positive Modality 5: Active Resolution Rather Than Mere Anti-Inflammation

Inflammation resolution is not equivalent to nonspecific suppression of inflammatory signaling. Specialized pro-resolving mediators (SPMs), including resolvins and maresins, actively promote efferocytosis, metabolic reprogramming, tissue repair, and restoration of homeostasis.

Human carotid plaque studies show lower resolvin D1 and lower SPM-to-leukotriene ratios in vulnerable plaque regions. In experimental atherosclerosis, RvD1 improves efferocytosis, reduces oxidative stress and necrosis, and increases fibrous-cap thickness.

Thus, a plaque-remodeling state should not merely be “less inflamed.” It should be **pro-resolving**.

Proposition P5. A beneficial clearance state requires active resolution and repair programs, not simply suppression of inflammatory markers.

6.6 Positive Modality 6: Structural Stabilization and Calcific Densification

The assumption that less vascular calcium is always better is not supported by imaging and clinical observational data. When calcium volume and density are separated, higher density can be associated with lower cardiovascular event risk [27, 28, 29, 30]. Statin therapy can reduce low-attenuation and non-calcified plaque while increasing calcified plaque or shifting calcium toward higher density.

This creates an important distinction:

- **microcalcification and low-density mineral in vulnerable regions** may increase local stress and risk;
- **dense macrocalcification** may accompany a more stable plaque phenotype.

Accordingly, “successful remodeling” cannot be defined as maximal reduction in CAC score.

Operationalizing structural stability. For CPRMH, “structural stability” is not a rhetorical endpoint. At minimum it should be operationalized against established plaque-pathology and imaging features. Thin-cap fibroatheroma (TCFA), a major precursor phenotype of plaque rupture, classically combines a large lipid/necrotic core with a macrophage-rich fibrous cap below approximately 65 μm in pathology and OCT-derived definitions [51, 52]. Consequently, a purported clearance intervention should not be called beneficial if it reduces one plaque material while producing cap thinning, VSMC loss, greater necrotic-core exposure, or increased matrix degradation. Candidate safety readouts therefore include fibrous-cap thickness where measurable, necrotic-core or low-attenuation plaque burden, macrophage-rich inflammatory features, collagen integrity, and evidence of plaque rupture, erosion, or calcified nodules.

Proposition P6. The clinically favorable endpoint may be a safer plaque architecture rather than minimal total calcium burden.

7. Negative Modalities

7.1 Negative Modality 1: Diffuse Acidification

Diffuse extracellular plaque acidity is not a benign version of osteoclast-mediated resorption. Human macrophage experiments show that reducing extracellular pH strongly impairs cholesterol efflux, especially ABCA1-mediated efflux, reduces expression of cholesterol-export machinery, and inhibits apoE secretion.

Acidic intimal fluid can also enhance LDL modification and retention, activate inflammasome-related inflammatory signaling, and increase proteolytic conditions.

Therefore:

The same chemical variable—proton activity—can be beneficial at a sealed mineral interface and harmful when distributed through the plaque.

This is the central spatial paradox of the framework.

7.2 Negative Modality 2: Incomplete Osteoclast-Like Resorption

Macrophages may possess CA2 but lack a complete resorptive phenotype. Human calcified plaque macrophages show impaired osteoclast-like differentiation and low cathepsin K activity. In this state, the plaque may contain acid-generating

potential without sufficient machinery for efficient mature-mineral resorption.

This negative modality predicts:

formation suppression > mineral removal

and therefore:

slower progression or stabilization without true decalcification.

7.3 Negative Modality 3: Protease Spillover

Osteoclast-like resorption requires matrix-processing enzymes. Yet collagen and elastin degradation in the wrong location can weaken the fibrous cap.

Macrophage-derived cathepsins and matrix metalloproteinases can participate in extracellular matrix degradation within atherosclerotic plaque. Therefore an intervention that indiscriminately increases “resorptive” macrophage activity could improve mineral dissolution while simultaneously increasing rupture risk.

The framework therefore imposes a structural constraint:

Mineral-facing resorption must not become cap-facing proteolysis.

7.4 Negative Modality 4: Foam-Cell Export Failure

Candidate metabolic signals can work differently before and after foam-cell transformation.

A useful example is GPR109A. In preclinical atherosclerosis, BHB-related signaling through GPR109A has been linked with anti-inflammatory effects and increased cholesterol-efflux machinery. Yet human macrophage-derived foam cells down-regulate GPR109A, and niacin fails to increase cholesterol efflux or key reverse-cholesterol-transport genes after foam-cell transformation.

This creates a strong boundary condition:

A state-setting signal cannot be assumed to remain effective after plaque cells undergo advanced phenotypic transformation.

7.5 Negative Modality 5: Defective Efferocytosis and Clearance Overload

Advanced plaques already have impaired clearance. Increased lipid hydrolysis, macrophage death, or mineral-associated injury could increase debris generation faster than efferocytosis can handle it.

MerTK cleavage, CD47-mediated anti-phagocytic signaling, metabolic dysfunction, and lysosomal stress can all reduce clearance capacity [15, 16]. In this state, attempts to “mobilize” plaque contents may increase necrotic burden rather than reduce it.

7.6 Negative Modality 6: Mature-Mineral Persistence

Mature vascular apatite can be highly persistent. Experimental correction of a pro-calcific environment does not necessarily lead to complete regression of established mineral.

This yields a critical epistemic rule:

Evidence that formation decreases cannot be reported as evidence that mature calcium is removed.

The framework therefore separates four outcomes:

1. reduced initiation;
2. slower propagation;
3. stabilization or phase transformation;
4. true regression/removal.

Each requires different evidence.

7.7 Negative Modality 7: Fasting Context Failure

Fasting is not uniformly anti-atherosclerotic.

In LDLR-deficient mice, some intermittent-fasting protocols reduced atherosclerosis, macrophage accumulation, necrotic-core area, and increased cap thickness and collagen. Yet other fasting regimens worsened dyslipidemia, insulin resistance, and atherosclerosis in hypercholesterolemic mice. More recent time-restricted feeding experiments found benefit in LDLR-knockout mice but not ApoE-knockout mice, implicating downstream hepatic lipoprotein clearance as a necessary condition.

This means:

The same fasting schedule can enter different modalities depending on genetic, metabolic, and lipoprotein-clearance context.

Fasting is therefore treated as a **state setter**, not a positive modality by definition.

Context-Lock Rule. Context dependence cannot be used as an unlimited post-hoc rescue. Any genetic, metabolic, hepatic-clearance, renal, medication, sleep, or circadian variable that is allowed to invoke Negative Modality 7 must be declared **before outcome analysis**, together with its operational definition and the direction in which it is expected to modify the state-setting effect.

A null result obtained inside a pre-specified eligible context counts as negative evidence against the state-setting proposition. A newly invented, unmeasured, or retrospectively selected context cannot rescue that null result unless the modifier is independently validated in a separate dataset.

This rule separates legitimate biological heterogeneity from theory-protective redefinition after the result is known.

7.8 Negative Modality 8: Ketosis–ApoB Inflow Conflict

A fasting- or carbohydrate-restriction strategy can improve glycemia, body weight, triglycerides, or BHB exposure while simultaneously increasing LDL-C and apoB in some individuals [56, 57]. Randomized and meta-analytic evidence confirms heterogeneous lipid responses to ketogenic or very-low-carbohydrate diets, including clinically substantial LDL-C/apoB elevation in a subset of lean, metabolically healthy participants (PMIDs: 33801247, 36931263, 40935153). This creates a direct conflict inside CPRMH’s own hydraulic principle: an upstream state setter that suppresses mineral-formation signaling could still worsen the lipid-inflow side of the system.

The counter-evidence must also be retained. In the selected KETO-CTA cohort of lean ketogenic-diet hyper-responders, one-year plaque progression was associated more strongly with baseline plaque than with apoB exposure [58]. That study does not establish long-term safety of severe apoB elevation, but it prevents the framework from treating diet-induced apoB elevation as a proven short-term plaque accelerator in every phenotype.

The CPRMH interpretation is therefore conditional:

Ketosis is not intrinsically positive or negative. Its net modality depends partly on whether any benefit in metabolic switching or mineral-formation suppression is outweighed by increased apoB-containing lipoprotein inflow.

This is one reason the grassroots cycle proposed here is **not a ketogenic-diet prescription**. It uses sleep-spanning

fasting and relative starch timing while retaining minimally processed carbohydrate sources and monitoring the broader inflow–clearance balance.

7.9 Negative Modality 9: Sleep-Anchor Instability

SAGDRC requires a temporal reference point. If the onset and duration of the principal sleep episode vary markedly from cycle to cycle, the relative gates can remain internally correct while repeatedly shifting against endogenous circadian phase. In that situation, “recalculate S” can become a faithful tracking system for the very instability that may be biologically adverse.

Sleep irregularity is independently associated with incident cardiovascular events and with subclinical atherosclerosis in the Multi-Ethnic Study of Atherosclerosis (MESA), including higher coronary calcium burden among participants with greater variability in sleep duration or timing [74, 75]. These observational findings do not prove that stabilizing sleep timing reverses plaque, but they make temporal stability a plausible modifier rather than a mere convenience.

CPRMH therefore distinguishes **sleep-anchor position** from **sleep-anchor stability**:

A stable or partially overlapping sleep anchor is proposed as Level 0 of SAGDRC. Relative feeding and stimulant gates are layered onto that anchor; they should not be assumed to neutralize rapidly rotating sleep timing.

For rotating schedules in which a single stable sleep episode is impossible, the framework proposes testing a **partial anchor window**: preserve as much overlap in the main sleep interval as real work and caregiving constraints allow, while treating residual circadian misalignment as an exposure to measure rather than a failure of adherence. This is an empirical hypothesis, not an established treatment rule.

7.10 Negative-Modality Lock and No-New-Blocker Rescue Rule

Negative modalities are biologically useful only if they take empirical risk. For a given plaque stage and causal contrast, the protocol therefore declares a blocker set $B_s \subseteq \{N1, \dots, N9\}$ before outcome access. Each included negative modality must have an operational variable, a threshold / margin or directional decision rule, a measurement window and a declared causal role. A negative modality that is not in B_s cannot be invoked after analysis to convert a failed prediction into a contextual success.

No-New-Blocker Rescue Rule: a previously unmeasured or newly proposed negative modality discovered after outcome access is hypothesis-generating only. It may enter the blocker set of a subsequent version, but it cannot rescue the current dataset. If a critical pre-specified blocker is technically unmeasurable, the result is blocker-indeterminate rather than supportive.

This rule extends the logic of the N7 Context-Lock Rule to the full negative-modality layer and makes the architecture symmetrical: both enabling dependencies and claimed failure states are prospectively exposed to defeat.

7.11 Measurement-Lock, Threshold-Robustness, and Causal-Precedence Rules

A falsifiable architecture also requires that the *means of observing failure* be exposed to risk. Otherwise an exhaustion failure could always be re-described as “the assay missed the true state”

or “the biologically correct threshold was slightly different.” CPRMH therefore freezes M_s and T_s before outcome access.

Measurement-Lock Rule: once a pre-specified measurement passes its own adequacy and quality-control criteria, post-hoc doubt about that measurement cannot rescue a failed prediction in the same dataset. Independent evidence of assay failure may invalidate the test, but it makes the result indeterminate rather than supportive.

Threshold-Robustness Rule: a result that supports CPRMH only at a narrow, post-hoc-favorable threshold is threshold-fragile evidence. Strong support requires the qualitative causal adjudication to remain stable across the prospectively specified plausible threshold range, or else the threshold uncertainty must be carried explicitly into the conclusion.

Causal-Precedence Rule: a negative modality cannot be operationalized by the same downstream outcome it is invoked to explain. The blocker signal must be measured upstream, earlier, or through a distinct construct with a declared causal role.

8. Allied Modalities

8.1 ApoB and LDL reduction: reducing inflow before increasing outflow

The strongest human precedent for safe plaque remodeling is intensive lipid lowering [49, 68, 69]. Serial intravascular and CT imaging studies show reductions in non-calcified or low-attenuation plaque under statin-based therapy, even as calcific components may increase [31, 32, 33, 34].

Conceptually, apoB lowering reduces **material inflow**, making it easier for efflux, efferocytosis, and structural repair to produce net regression.

This is the hydraulic principle of the framework:

A clearance system is unlikely to empty a compartment efficiently while pathological inflow remains high.

The strongest direct human benchmark is not fasting but intensive apoB/LDL lowering. In the GLAGOV randomized trial, statin-treated patients receiving evolocumab achieved a mean time-weighted LDL-C of approximately 36.6 mg/dL and showed a reduction in percent atheroma volume, with plaque regression occurring in a greater proportion of participants than with placebo [49]. A plaque-composition analysis found regression of overall atheroma burden without a significant differential reduction in calcium volume [50]. This is highly informative for CPRMH: **safe regression of plaque burden is biologically possible in humans, but calcium need not disappear in parallel.**

8.2 Pyrophosphate and other mineral-growth brakes

Extracellular PPI inhibits hydroxyapatite formation and propagation [4, 53, 54, 55]. PPI-related systems therefore support formation suppression even if they do not resorb established mineral.

The human-genetic evidence is unusually strong for a mechanistic framework. Loss-of-function variants in **ENPP1**, which generates extracellular PPI from ATP, cause generalized arterial calcification of infancy (GACI); pathogenic variants in **ABCC6** can also reduce extracellular PPI and produce overlapping GACI/PXE phenotypes. Recent reviews describe the shared $ABCC6 \rightarrow \text{extracellular ATP} \rightarrow \text{ENPP1} \rightarrow \text{PPI}$ pathway and the severe arterial calcification that follows disruption of this inhibitory system (PMIDs: 39343469, 42188070, 38392293).

Table 5. Measurement and threshold discipline for prospective CPRMH tests

Locked object	Minimum pre-specified content	What is not admissible after outcome access
M_s : construct / assay	biological construct, specimen / imaging domain, instrument or assay, spatial and temporal scale, available construct-validity evidence	replacing the failed construct with a different biomarker because the original result is inconvenient
M_s : adequacy	precision / repeatability target, quality-control limits, missingness rule, assay-failure rule, minimum signal-to-noise or measurement adequacy needed for adjudication	declaring an assay “probably insensitive” after it has passed its locked QC / adequacy criteria, absent independent evidence of failure
T_s : decision threshold	target-engagement, blocker, responder or harm threshold justified before analysis, plus a plausible biological range	moving the cutoff after examining outcomes, or reporting only the cutoff at which the architecture survives
T_s : robustness	sensitivity analysis across the locked plausible range and explicit classification of threshold-stable versus threshold-fragile conclusions	treating a conclusion that survives only inside a narrow data-fitted cutoff as confirmation
Causal precedence	blocker / mediator measure occurs upstream of, or is analytically separable from, the downstream endpoint it is invoked to explain	defining N5 by necrotic-core failure and then using the same necrotic-core failure as evidence that N5 was active

These are human natural experiments demonstrating causal sufficiency in the opposite direction: **remove a major endogenous anti-mineralization brake and arterial mineralization can become catastrophic.**

The chemistry provides a further triangulation. PPi contains a hydrolysable P–O–P bridge and inhibits hydroxyapatite crystal propagation. Bisphosphonates replace the central oxygen with carbon, producing a hydrolysis-resistant P–C–P motif. This does not make bisphosphonates a general vascular-decalcification therapy, but it illustrates how endogenous mineral-regulatory chemistry can be pharmacologically stabilized.

This modality is allied because it changes the **mineral balance boundary condition**.

8.3 Vitamin-K-dependent MGP

Matrix Gla protein is an endogenous inhibitor of vascular calcification whose activation depends on vitamin K [4]. Clinical evidence is more consistent for changes in biochemical markers and possible slowing of progression than for established mineral regression.

MGP should therefore be conceptualized as an **anti-new-mineral ally**, not a decalcifying agent.

8.4 Autophagy and lipophagy

Autophagy potentially links several modules:

- BHB-related suppression of VSMC calcification;
- macrophage organelle quality control;
- lipophagic mobilization of stored lipid;
- cholesterol availability for efflux.

Because excessive or context-dependent autophagy can have different effects across cell types and disease stages, it is an allied modality rather than a guaranteed positive mechanism.

8.5 Specialized pro-resolving mediator substrate availability

EPA and DHA are precursors for multiple SPM families [17, 18]. Diets rich in these fatty acids can alter the substrate environment for resolution biology, but substrate intake is not equivalent to local SPM production within plaque.

The relevant causal chain is therefore:

dietary precursor → absorption → enzymatic conversion → local SPM availability → macrophage resolution/efferocytosis → plaque phenotype

Every link requires verification.

8.6 Exercise as an independent state setter

Exercise changes insulin sensitivity, substrate use, inflammatory signaling, lipoprotein metabolism, and—under some conditions—ketone production. It provides a valuable comparator for fasting.

If fasting and exercise produce similar downstream plaque-relevant changes when BHB exposure overlaps, BHB-mediated mechanisms become more plausible. If effects diverge, fasting-specific pathways or non-ketone mechanisms become more likely.

8.7 Existing pharmacology as mechanistic triangulation—and as evidence of white space

Existing drug development helps test whether the modality architecture corresponds to biologically separable functions.

- **Mineral-growth suppression:** PPi analog chemistry, bisphosphonate classes, and investigational hydroxyapatite-growth inhibitors demonstrate that mineral initiation/growth can be targeted without systemic acidification. These approaches should not be conflated with removal of mature vascular mineral.
- **State-setting / calcification suppression:** SGLT2 inhibitors provide existing pharmacologic entry points into calcification-suppression pathways, including SIRT1/HIF-1 α and NLRP3 signaling, independent of a simple “BHB does everything” explanation.
- **Lipid export:** apoA-I/reconstituted-HDL approaches act on the cholesterol-acceptor side of ABCA1-mediated efflux. Cyclodextrin chemistry provides a proof of principle that cholesterol can be handled through molecular encapsulation rather than bulk mechanical detachment, although it is not an established atherosclerosis treatment.
- **Failure of single-axis optimization:** systemic LXR agonism can increase ABCA1/ABCG1 expression yet produce hepatic lipogenesis and hypertriglyceridemia; historical CETP experience similarly warns that improving HDL-related biomarkers is not equivalent to improving net macrophage cholesterol export or clinical outcomes.

- **True white spaces:** no approved therapy currently provides selective macrophage-mediated, mineral-facing, spatially confined vascular demineralization, and no approved cardiovascular therapy specifically restores plaque efferocytosis as its primary mechanism. CD47–SIRP α blockade is a cross-domain mechanistic candidate from oncology, but systemic disruption of “don’t-eat-me” signaling carries obvious hematologic safety constraints.

These examples support the coordination claim: **moving one modality can improve one axis while worsening another.**

9. Sleep-Anchored Everyday State-Setting and the Grassroots Daily Remodeling Cycle

The framework deliberately separates **state-setting exposure** from **plaque-remodeling mechanism**. It also separates **biological timing** from **social clock time**.

A fixed rule such as “stop carbohydrate at 14:00” presumes a conventional day schedule. That rule becomes incoherent for a nurse sleeping at 09:00 after a night shift, a transport worker beginning work before dawn, or a caregiver whose main sleep episode occurs in two different blocks. A framework intended for health equity cannot define healthy timing solely by reference to the timetable of daytime salaried work.

The proposed solution is a **sleep–wake-anchored relative-time architecture**.

Let *S* denote the intended onset of the principal sleep episode and *W* the end of that sleep episode. The interval from *W* to the next *S* is the person’s practical waking day. Behavioral gates are positioned relative to *S* and *W* rather than assigned universal wall-clock times.

9.1 The Sleep-Anchored Grassroots Daily Remodeling Cycle (SAGDRC)

SAGDRC is a translational sub-hypothesis of CPRMH. It proposes that recurrent daily coordination of feeding, movement, stimulant withdrawal, fasting, and sleep may increase the duration or frequency of remodeling-permissive states without requiring expensive products, gym membership, or standardized work schedules.

The cycle begins with a **Level 0 temporal condition** and then adds four temporal gates and two movement levels. Level 0 is not a demand for a conventional daytime schedule. It is the attempt to preserve the most stable feasible principal-sleep or partial-anchor window within the person’s actual work and caregiving constraints. The theory therefore separates **stability of the anchor** from **where the anchor sits on the clock**.

Gate 1 — Caffeine withdrawal: *approximately S–8 to S–9 hours.* A 2023 systematic review and meta-analysis found that caffeine reduces total sleep time and deep sleep and estimated that a standard coffee should be consumed at least approximately 8.8 hours before bedtime to avoid a measurable reduction in total sleep time [36]. The practical CPRMH rule is therefore relative rather than clock-based. Because TRE does not uniformly improve sleep, sleep quality remains an outcome to measure rather than an assumed benefit [46]:

Stop caffeine roughly 8–9 hours before the intended main sleep episode.

This gate is intended to protect the **sleep/repair arm** of the framework rather than to induce ketosis.

Gate 2 — Starch-reduction gate: *proposed at approximately S–4 to S–6 hours.* The framework proposes that starch-dominant foods should be concentrated earlier in the waking episode and reduced during the final several hours before the principal sleep episode. The initial empirical range proposed for testing is **4–6 hours before S**.

This is deliberately framed as a **new parameter to be tested**, not an established universal threshold. Its mechanistic rationale is to reduce late-waking-period glycemic and insulin load, facilitate a progressive transition toward lower substrate availability, and reduce the probability that the final pre-sleep period remains dominated by repeated starch-rich feeding.

The rule is not “no carbohydrate ever after this point.” It is a food-structure rule: after the starch gate, meals preferentially shift toward vegetables, legumes/soy foods, eggs, fish, and other minimally processed protein sources, with starch-dominant foods reduced.

Gate 3 — Last-calorie gate: *approximately S–2 to S–3 hours.* The proposed low-cost implementation ends meaningful caloric intake approximately **2–3 hours before the main sleep episode**, when feasible. This separates digestive load from the beginning of the principal repair/sleep interval and creates an immediate pre-sleep fasting segment.

Gate 4 — Sleep-spanning fasting window. Fasting begins after the final caloric intake, continues through the principal sleep episode, and can extend into the next waking period before the first meal. For many conventional schedules this can generate approximately **12–14 hours of total fasting without skipping an entire waking day**; however, the theoretical variable is not a fixed fasting duration alone.

The exposure of interest is:

fasting duration × achieved BHB exposure × recurrence × sleep quality × feeding quality before and after the fast.

Thus, a 12-hour fast with lower apoB-generating intake, adequate sleep, and repeated daily movement is not assumed to be equivalent to a 12-hour fast surrounded by ultraprocessed food, sleep deprivation, or large late meals.

9.2 Why sleep and wake are the primary practical anchors

Sleep and wake have three advantages over fixed clock time.

First, they are observable without laboratory testing. Second, they make the model usable by workers outside conventional daytime schedules. Third, they force the theory to treat **sleep as an active repair interval**, not merely as dead time between dietary interventions.

This does not imply that endogenous circadian phase is irrelevant. Night-shift experiments show that eating during biological night can impair glucose tolerance and that restricting or redistributing food intake during simulated night work can mitigate metabolic and cardiovascular disturbances [37, 38, 39, 40, 59]. Sleep-relative timing is therefore a **pragmatic field proxy**, not a claim that circadian biology can be reduced to sleep time.

The framework consequently adopts a **two-level temporal hierarchy**:

Level 1 — practical anchor: schedule caffeine withdrawal, starch reduction, last calories, and sleep-spanning fasting relative to the person’s main sleep episode.

Level 2 — circadian safety modifier: when a person is awake through the biological night, avoid large or starch-heavy feeding

during that interval when feasible, even if sleep-relative gates alone would permit it.

This explicitly prevents the false equivalence “**sleep time = circadian phase.**” The sleep anchor is proposed because it is practical and portable; endogenous circadian phase remains a biologically stronger candidate explanation that can supersede the sleep-relative rule if future experiments show it consistently predicts relevant outcomes better.

9.3 Application to night-shift and non-standard workers

A night-shift worker should not be instructed to stop starch at 14:00 simply because a daytime worker does. Instead, the same sequence is shifted around the worker’s principal sleep episode.

If a worker intends to sleep at 09:00, for example, the relative architecture places caffeine withdrawal roughly around midnight to 01:00, the proposed starch-reduction gate roughly around 03:00–05:00, and the last-calorie gate roughly around 06:00–07:00. Fasting then spans the main daytime sleep period. These times are illustrations generated from the relative rules, not universal prescriptions.

For rotating shifts, the framework no longer treats full re-anchoring to each upcoming sleep episode as the preferred default. Instead it uses a hierarchy: **first preserve the most stable feasible sleep or partial-anchor window; second position relative gates around that anchor; third, when work makes the anchor move, quantify the degree of displacement and circadian misalignment rather than silently redefining them away.** Full cycle-by-cycle recalculation remains a comparator strategy, not the assumed optimal strategy. This revision creates a direct test of Negative Modality 9: whether sleep-anchor stability adds biological value beyond total sleep duration and total energy intake.

Evidence in shift workers is still limited. A randomized trial in firefighters working 24-hour shifts found that a self-selected approximately 10-hour eating window was feasible and improved selected cardiometabolic measures in higher-risk participants [37]. Experimental night-shift trials also suggest that fasting or restricting food during the biological night can attenuate glucose intolerance compared with consuming meals or snacks at night. A 2025 randomized simulated-night-work analysis further reported that daytime-only eating prevented adverse changes in several cardiovascular risk markers, including autonomic indices and PAI-1, despite circadian misalignment [59]. A 2026 systematic review identified only four randomized shift-work meal-timing trials and judged the evidence promising but still too sparse and methodologically limited for firm generalization [41].

These findings do not prove SAGDRC. They establish that **meal timing remains biologically consequential even when work schedules are abnormal**, making shift workers a critical falsification population rather than an exception to the theory. Importantly, **a stable sleep anchor is not the same claim as recommending permanent night work.** A 2026 umbrella review found different risk patterns across fixed-night and rotating schedules rather than a simple universal ranking [76]. SAGDRC therefore targets stability of the sleep opportunity within unavoidable work constraints; it does not infer that one employment schedule is globally safer.

9.4 Embedded Activity First: movement without creating a new time burden

The movement arm of SAGDRC is deliberately divided into two levels.

Level 1 — Embedded daily movement. The first goal is not to “find time to exercise.” It is to harvest **safe movement opportunities that already exist inside daily life:**

- walking for transport or errands;
- taking stairs where feasible and safe;
- house cleaning, sweeping, mopping, washing, gardening, and ordinary domestic movement;
- carrying groceries or household loads within safe capacity;
- walking while making phone calls or accompanying family members;
- short bouts of walking or light activity after meals;
- breaking prolonged sitting with brief movement;
- using naturally occurring movement opportunities during work **without deliberately increasing harmful workload, heat stress, repetitive strain, or recovery debt.**

WHO guidance recognizes transport, occupational activity, household tasks, and other everyday movement as physical activity [42]. But CPRMH does **not** infer that every form of occupational physical activity is cardioprotective. The occupational physical-activity paradox literature suggests that prolonged, repetitive, low-control or insufficiently recovered work activity can differ physiologically from leisure-time exercise and may be associated with less favorable cardiovascular outcomes in some occupations [64]. An individual-participant meta-analysis further found that leisure-time physical activity was associated with lower cardiovascular and all-cause mortality across most occupational-activity strata [65].

Therefore:

Embedded Activity First means harvesting low-cost, safe movement opportunities already present in daily life—not prescribing “more work” to people whose jobs are already physically demanding.

This level has the lowest potential combined **money cost + travel cost + equipment cost + additional-time cost**, but its benefit must still be evaluated against fatigue, injury, heat, recovery, and occupational context.

Level 2 — Added intentional activity. Where time, physical capacity, and local conditions permit, individuals can add brisk walking; cycling; running or other aerobic activity; body-weight resistance exercise; stair or step training; and community sport or other structured activity.

The evidence-based population target remains the WHO recommendation of 150–300 minutes of moderate aerobic activity weekly, or an equivalent vigorous amount, plus muscle-strengthening activity on at least two days per week. SAGDRC does not redefine these targets. It changes the **access strategy:** safe embedded activity comes first, and planned exercise is an additional layer rather than the only legitimate route to movement.

9.5 A low-cost Thai feeding matrix

The food arm is designed around ordinary Thai foods rather than imported “health foods.” The guiding principles are consistent with contemporary cardiovascular and vascular-calcification dietary evidence and with the 2026 American

Heart Association statement and Thailand's 2025 Food-Based Dietary Guidelines [35, 43, 44]: vegetables and fruit, whole or less-refined grains, legumes, fish, eggs and other minimally processed protein sources, unsaturated rather than saturated fat, and lower intake of added sugar, sodium, and ultraprocessed foods.

A low-cost Thai implementation can therefore be built from foods such as:

- rice mixed with brown or less-polished rice when affordable, rather than requiring specialty grains;
- local vegetables such as morning glory, Chinese kale, cabbage, cucumber, pumpkin, gourds, eggplant, long beans, okra, and seasonal greens;
- mung beans, soybeans, tofu, unsalted peanuts, and other legumes;
- eggs;
- local fish such as mackerel and sardine-type fish rather than requiring salmon;
- seasonal whole fruit rather than juice;
- water as the default beverage.

The principal reduction targets are not culturally specific “bad foods” but **high-frequency pathological inflow**: sugar-sweetened beverages, repeated refined-starch snacking, heavily processed meat, ultraprocessed snacks, excessive sodium, and recurrent late-waking-period starch-heavy meals.

The Thai Department of Health's updated 2025 Food-Based Dietary Guidelines emphasize local and seasonal foods, unpolished grains, vegetables and fruits, fish, eggs, legumes, and minimizing sugar, fat, and salt [44]. This alignment is important because the grassroots framework should not create a new financial barrier by making cardiovascular health depend on imported products or supplements.

The matrix is intentionally **functional rather than prestige-based** (Table 6):

The design principle is **substitution before supplementation**: replace repeated sugar-sweetened drinks, processed meats, ultraprocessed snacks, excessive sodium, and late starch-heavy eating before asking households to purchase specialty products.

9.6 Metabolic–Renal Adapted Branch: obesity, type 2 diabetes, and CKD G3

Obesity, type 2 diabetes, and CKD G3 are **not exclusion criteria** for CPRMH. They are a high-load phenotype in which multiple parts of the architecture are simultaneously stressed: apoB and glycemic inflow, renal phosphate and acid–base handling, VSMC calcification pressure, medication effects, sleep disruption, physical-capacity constraints, and vulnerability to dehydration or hypoglycemia. Excluding this phenotype would weaken both the biological test and the equity claim.

Current KDIGO guidance explicitly recommends physical activity for people with CKD to the level compatible with cardiovascular and physical tolerance, discourages sedentary behavior, encourages weight loss in people with obesity and CKD, and recommends healthy diverse diets emphasizing plant-based foods and reduced ultraprocessed foods [71]. For CKD G3–G5, KDIGO and the 2026 ADA Standards recommend protein intake around 0.8 g/kg/day rather than high-protein dieting, with sodium, potassium, phosphorus, and other dietary adaptations individualized to kidney function, comorbidities,

medication use, and laboratory values [71, 72]. ADA likewise emphasizes individualized nutrition and weight management in diabetes and obesity rather than a single macronutrient formula [73].

The adapted branch therefore changes **priority and dose**, not the underlying theory:

The important theoretical move is that **the high-risk phenotype stays inside the model**. CKD and diabetes change the transition probabilities among modalities; they do not make coordinated remodeling conceptually irrelevant. Indeed, this phenotype is a stronger stress test: if CPRMH only functions in metabolically healthy people with normal kidney function, its population and equity claims must be sharply reduced.

For obesity + type 2 diabetes + CKD G3, SAGDRC tests temporal organization and low-cost exposure under renal-metabolic constraints; it does not require prolonged fasting, ketogenic eating, high protein intake, or unsupervised medication adjustment.

9.7 Relative daily sequence

Table 8 states the sleep-relative sequence in full.

The table is intentionally **relative**. A person sleeping at 22:00 and a person sleeping at 09:00 use the same architecture with different wall-clock times.

9.8 The novel temporal proposition

The new proposition is not that any one element—fasting, walking, low-starch evening meals, caffeine withdrawal, or sleep—independently “cleans” plaque.

It is that:

Repeated coordination of low-cost behaviors around the individual's sleep–wake cycle may create recurrent remodeling-permissive windows in which pathological inflow and new mineral formation fall while lipid handling, autophagic processes, efferocytosis, resolution, and structural repair become relatively more favorable. Repetition of these windows may cumulatively shift plaque dynamics from accumulation-dominant toward stabilization- and, under narrower permissive conditions, regression-compatible states.

This is a new synthesis claim. Its novelty lies in **temporal coordination, relative-time anchoring, material-specific clearance, and feasibility**, not in asserting that walking, sleep, vegetables, or fasting were individually unknown before this article.

10. Fasting, BHB and the New BMAL1–MERTK Evidence: A State-Setting Case Study

10.1 What was already known

BHB suppresses vascular calcification in multiple preclinical systems through autophagy, HDAC9–NF- κ B and AMPK/Nrf2-related mechanisms [19, 20, 21]. In a large non-diabetic human cohort, fasting ketonuria was inversely associated with prevalent CAC and modestly associated with slower CAC progression [22]. Earlier mouse studies showed that intermittent or time-restricted feeding can reduce lesion burden or improve plaque features in some models, but not all; the response differs by genetic and lipoprotein-clearance context [23, 24, 25, 26].

Table 6. Low-cost Thai feeding matrix organized by function rather than prestige

Low-cost Thai food function	Examples	CPRMH relevance
plant/fiber matrix	local vegetables, gourds, leafy greens, legumes	supports lower energy density and replaces ultraprocessed inflow
affordable protein	eggs, tofu, soy, mung beans, local fish	supports satiety and muscle maintenance without requiring premium meat
whole/less-refined starch earlier in waking period	brown/unpolished rice when feasible, mixed rice, minimally processed tubers	preserves culturally realistic carbohydrate intake while allowing relative starch timing
unsaturated/SPM-precursor sources	local oily fish, peanuts and other unsalted nuts/seeds as affordable	allied lipid/resolution substrate, not a direct plaque solvent
default hydration	water	removes sugar-sweetened-beverage inflow at minimal cost

Table 7. Metabolic–renal adaptation of SAGDRC for obesity + type 2 diabetes + CKD G3

Domain	Standard SAGDRC concept	Metabolic–renal adaptation
Sleep / timing	sleep-relative gates	preserve the most stable feasible sleep anchor; sleep irregularity and shift displacement become measured exposures
Fasting	sleep-spanning low-input interval, optional extension after waking	no mandatory prolonged fast ; duration must be compatible with glucose-lowering medication, glycemic stability, hydration, renal status, and clinical advice; the temporal-ordering hypothesis can be tested without long fasting
Starch / food quality	earlier starch allocation, minimally processed local foods	retain food-quality and timing logic, but avoid turning “lower starch” into a high-protein ketogenic prescription; emphasize affordable vegetables, legumes / soy, appropriate whole grains, and other foods compatible with individualized renal nutrition
Protein	ordinary minimally processed protein sources	CKD G3–G5 branch is anchored near guideline protein intake (about 0.8 g/kg/day) rather than high-protein dieting; frailty and nutritional status modify this target
Minerals / sodium	low-processed-food matrix	sodium reduction is prioritized; potassium and phosphorus are not universally restricted but individualized to laboratory values, renal function, medication use, and dietetic assessment
Movement	Embedded Activity First + added intentional activity	avoid sedentary exposure and accumulate tolerable activity; progress toward guideline activity according to cardiovascular, musculoskeletal, neuropathy, fall, and fatigue constraints
Weight	avoid pathological inflow	intentional gradual weight reduction is an allied modality in obesity where appropriate; rapid or unintentional loss remains a negative boundary
Monitoring in research	metabolic and plaque readouts	add eGFR, albuminuria, potassium, bicarbonate / acid–base status where available, blood pressure, glycemia / hypoglycemia exposure, medication class, hydration-related events, and CKD-mineral variables appropriate to disease stage

Table 8. Sleep-relative daily sequence for the SAGDRC implementation hypothesis

Relative phase	Low-cost action	Proposed CPRMH function
W to early waking period	First substantial meals; starch can be concentrated here; ordinary movement begins	fuel active period while limiting late accumulation pressure
Throughout waking period	Embed safe walking, transport and household movement; use work-related movement only where it does not add fatigue/recovery debt	glucose disposal, lipid handling, reduced sedentary exposure without occupational overload
S–8 to S–9 h	Stop caffeine	protect sleep/repair modality
S–4 to S–6 h	Reduce starch-dominant foods; shift toward vegetables + protein/legumes	lower late glycemic/substrate pressure; begin transition
S–2 to S–3 h	End meaningful calories when feasible	begin pre-sleep fasting interval
S through sleep	Continue fasting	low-input / sleep-repair interval
W to first meal	Optional fasting extension according to feasibility and safety	extend metabolic switching window
2+ days/week if feasible	Add strength work beyond embedded movement	increase metabolic reserve and muscle-mediated substrate disposal

10.2 The August 2026 evidence update: established plaque regression through BMAL1–MERTK

A study published on 11 August 2026 materially changes the evidence landscape. Feng and colleagues used a baseline-controlled Ldlr^{-/-} induction–intervention model and reported

that intermittent fasting reduced plaque size relative to both ad-libitum controls and a pre-intervention baseline, while improving stability features without additional endpoint lipid lowering [66]. Multi-omic and cellular experiments prioritized a macrophage BMAL1–MERTK axis: BMAL1 occupied the

tested Mertk promoter region, Mertk re-expression rescued impaired efferocytosis after Bmal1 knockdown, and MERTK inhibition reversed the benefit of BMAL1 overexpression [66].

This result is important for CPRMH for two opposite reasons. First, it gives independent mechanistic support to the proposition that a recurrent lifestyle-linked state can influence an established-plaque clearance modality through circadian and efferocytic machinery. Second, it **removes any novelty claim that “fasting activates efferocytosis” by itself**. The remaining novelty is broader: CPRMH predicts that this efferocytosis route must interact with lipid export, mineral handling, inflammatory resolution, inflow and structural repair before it can be generalized as safe coordinated remodeling.

Evidence update rule. New evidence that validates one CPRMH branch does not validate the whole architecture; it narrows uncertainty and shifts that branch from speculative synthesis toward supported mechanism while leaving the coordination claim independently falsifiable.

10.3 Dose-translation and mediator problems

Physiological BHB exposure during common 12–16-hour fasting is generally lower than the millimolar concentrations used in several mechanistic experiments. Fasting duration alone is therefore a poor exposure variable. Future work should characterize **BHB peak × duration × recurrence × tissue exposure** while measuring energy intake, hepatic lipid flux, VLDL secretion, insulin, glucose, fatty-acid flux, apoB and circadian phase.

The new BMAL1–MERTK result also strengthens the mediator problem: a fasting effect need not be a ketone effect. BHB, circadian transcription, caloric exposure, lipoprotein flux and immune-state changes may operate in parallel or sequentially. The 12 August 2026 cardiovascular chrononutrition review likewise emphasizes circadian alignment as a major candidate mechanism of timed feeding [67].

10.4 Advanced-plaque and translation problems

Advanced human foam cells can down-regulate candidate receptors and lose efficient cholesterol-export responses [13, 14]. Mature atheroma is also relatively persistent, so suppression of new calcification should not be rewritten as removal of old mineral. A biologically plausible common outcome therefore remains:

less new deposition + improved cellular clearance + greater structural stability, with limited mature-mineral regression.

The 2026 fasting-regression study is preclinical. It does not establish coronary plaque regression from fasting in humans, does not establish an optimal fasting schedule, and does not show that mineral resorption is required for its effect. Those distinctions are essential to the CPRMH evidence hierarchy.

10.5 Relative-time fasting proposition

CPRMH treats fasting as a **sleep-spanning state transition** rather than a universally fixed clock interval. The primary exposure should be recorded relative to the principal sleep episode, with separate documentation of endogenous circadian phase. This is a pragmatic field architecture, not a claim that sleep relative time supersedes circadian biology. If direct circadian-phase measures consistently outperform sleep-relative timing, SAGDRC must be revised toward a stronger circadian anchor [38, 59, 67].

11. The Integrated Modality Architecture

The integrated architecture is represented as a causal routing problem rather than as a single pathway. Figure 1 separates recurrent state-setting exposures and background modifiers from six material-specific remodeling tasks, then imposes a coordination gate before classifying downstream plaque trajectories. The diagram is deliberately acyclic: repeated daily exposure is represented as repeated traversal of the same analytical cycle rather than as a feedback arrow inside the graph.

Figure 2 translates the same logic into one practical sleep–wake cycle. It is drawn as a linear one-cycle sequence rather than a closed loop so that the temporal ordering remains visible and the figure does not imply that any gate is already proven to cause plaque regression.

The two figures separate the theoretical claim from its practical translation: Figure 1 is the mechanistic architecture; Figure 2 is a low-cost temporal implementation hypothesis that must be tested against fixed clock time and measured circadian phase.

12. Positive, Negative, and Allied Modality Matrix

Table 9 arranges the three modality classes against the domains in which they act, so that a positive modality is never read in isolation from the negative modality that can defeat it or the allied modality that conditions it.

13. Pre-Specified Outcome Adjudication

A coordination theory becomes non-falsifiable if every mixed result can be re-described as “remodeling.” CPRMH therefore requires a study-level adjudication rule **before outcomes are inspected**. Each empirical test must pre-specify (i) the target population and causal contrast, (ii) one primary plaque-burden or material endpoint appropriate to the study, (iii) a set of critical structural-vulnerability / safety endpoints, (iv) the analysis window and treatment of intercurrent events, and (v) clinically or biologically justified change and harm margins. This follows the general logic of prospectively defining the target estimand and interpretation before analysis rather than allowing the observed data to redefine the question [79]. Biomarker improvement alone cannot establish plaque regression.

The primary adjudication rule is asymmetric: improvement in one material axis cannot rescue worsening of a pre-specified structural-safety axis. Conversely, stable or increasing calcific density does not by itself negate benefit when overall burden and vulnerability satisfy the pre-specified regression or stabilization rule.

This hierarchy prevents retrospective endpoint switching. It also distinguishes the clinically relevant claim of *safe* remodeling from the weaker observation that one biomarker or one plaque constituent changed.

13.1 Safety gate: confidence intervals, harm margins and multiplicity

“No statistically significant worsening” is **not** accepted as evidence of safety. For every critical safety endpoint, the protocol must define a biologically or clinically justified harm margin before analysis and use the confidence interval for the treatment contrast to exclude deterioration beyond that margin. The logic borrows the discipline of pre-specified non-inferiority margins and confidence-bound interpretation,

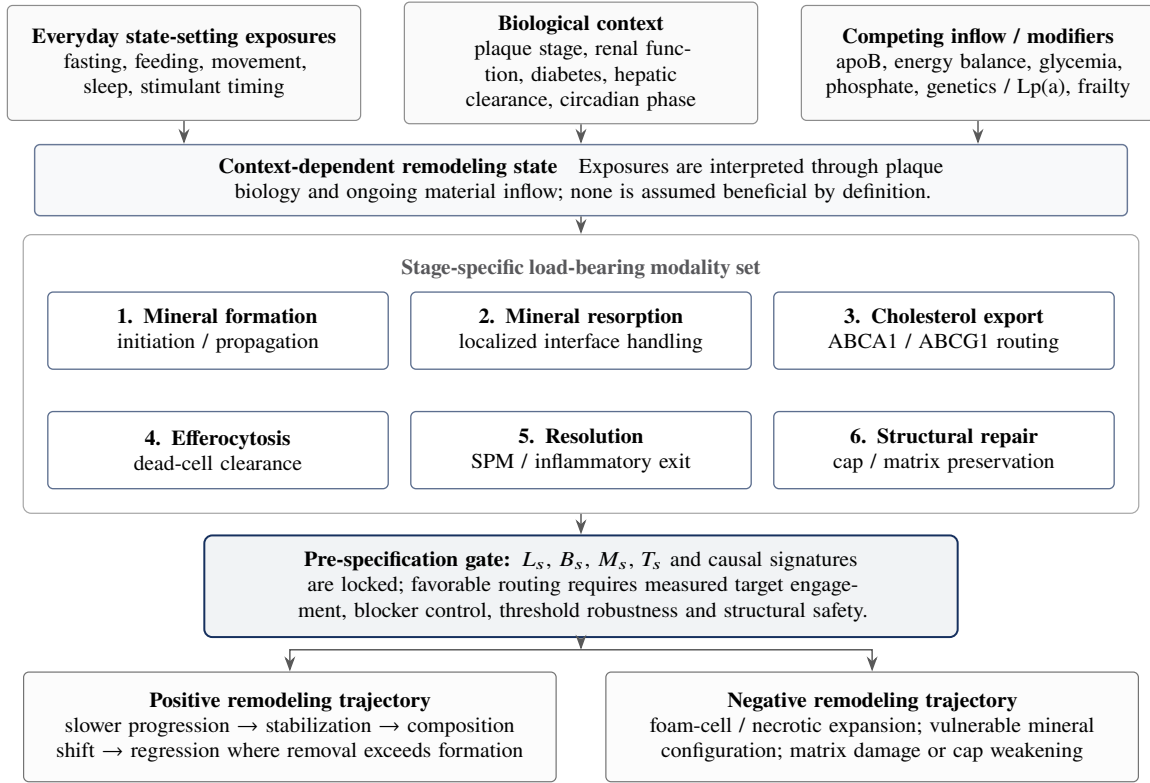


Figure 1. CPRMH causal architecture. Upstream state-setting exposures, conditioning biology and competing inflow are separated from the material-specific remodeling modalities. For each plaque stage, the load-bearing set L_s , negative-modality blocker set B_s , and causal signatures are locked prospectively rather than deleted, added, or re-labeled after results are known. The modalities may be arranged through parallel conditional dependency or serial causal dependency; the downstream dual-lock gate asks whether target engagement and the pre-specified dependencies coexist with blocker control and structural safety. This routing avoids implying that fasting, feeding or exercise directly “cleans” plaque.

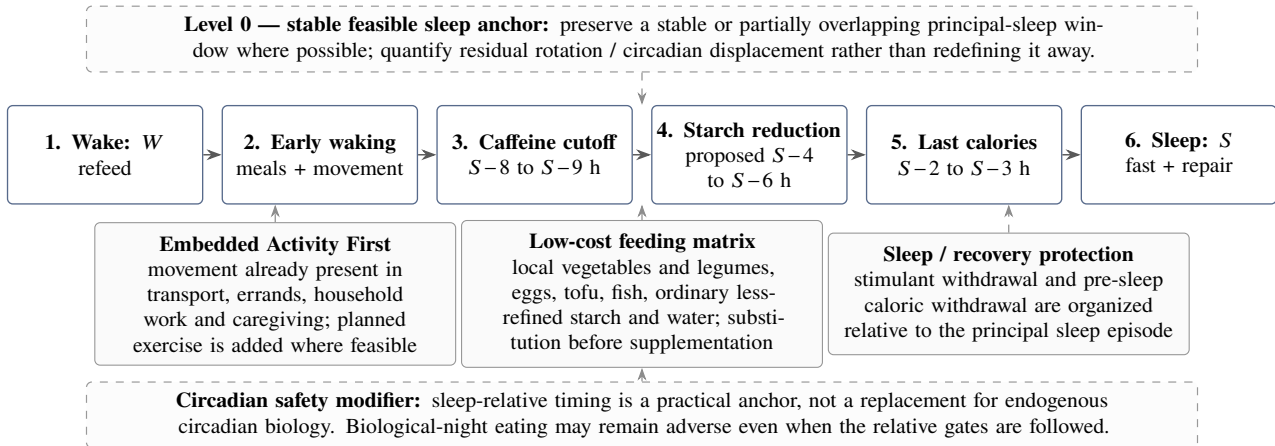


Figure 2. Sleep-Anchored Grassroots Daily Remodeling Cycle (SAGDRC), one-cycle representation. Level 0 distinguishes stability of the sleep anchor from its clock position. S denotes onset of the principal sleep episode and W waking from it. The caffeine cutoff is evidence-informed for sleep protection; the $S-4$ to $S-6$ hour starch-reduction interval is a CPRMH hypothesis parameter; the $S-2$ to $S-3$ hour last-calorie interval is a pragmatic test range. The lower row shows implementation allies rather than additional causal steps. In CKD / diabetes, fasting and food composition use the metabolic–renal adapted branch rather than identical dose assumptions.

without claiming that every CPRMH study is a regulatory non-inferiority trial [81]. If no defensible harm margin exists for a critical endpoint, the result cannot receive the strongest “safe regression” label; it remains safety-indeterminate until the endpoint can be adjudicated.

When several critical safety endpoints define structural integrity, the safety gate is **conjunctive**: every designated critical endpoint must pass its margin. Exploratory secondary endpoints may remain descriptive, but the primary classification cannot cherry-pick among them. Multiplicity and

endpoint hierarchy must be addressed prospectively in keeping with established multiple-endpoint principles [80]. This rule is intentionally conservative: absence of a harm signal is weaker evidence than exclusion of a pre-specified unacceptable harm.

14. Falsification Architecture

A conceptual model is useful only if observations can weaken or reject it.

Table 9. Positive, negative, and allied modalities within CPRMH

Domain	Positive modality	Negative modality	Allied modality
Mineral formation	lower VSMC osteogenesis / nucleation	phosphate-driven or inflammatory mineralization	PPi, MGP, Mg-related mineral regulation, BHB-autophagy pathways
Mineral removal	confined CA2/V-ATPase-like resorption	diffuse acidification; incomplete osteoclast program	anti-inflammatory state; mineral-interface targeting
Lipid handling	ABCA1/ABCG1 molecular efflux	free-cholesterol overload; transporter degradation	apoB lowering; HDL acceptor capacity; lipophagy
Cellular debris	efficient efferocytosis	MerTK cleavage; CD47; secondary necrosis	SPMs; metabolic support of efferocytes
Inflammation	active resolution	chronic inflammasome/protease signaling	EPA/DHA-derived SPM pathways
Structure	collagen maintenance / cap thickening	MMP/cathepsin spillover; SMC loss	lipid lowering; pro-resolving signaling
Calcific phenotype	stable dense microcalcification where appropriate	mechanically hazardous microcalcification	statin-associated compositional shift
Daily state	permissive metabolic switching	maladaptive fasting; hyperlipidemic rebound	TRE, exercise, meal composition under defined contexts
Temporal organization	stable or partially overlapping sleep anchor	N9: anchor instability ; repeated circadian displacement	feasible work scheduling, light timing, circadian-phase measurement
Renal–metabolic context	adapted low-input, dietary and activity exposures compatible with CKD / diabetes care	hypoglycemia, dehydration, high-protein or electrolyte-inappropriate translation	renal monitoring, medication-aware design, individualized nutrition

Table 10. Theory-level adjudication of plaque outcomes

Adjudication class	Minimum pattern	Interpretation for CPRMH
Safe regression	pre-specified plaque / atheroma burden decreases and <i>every critical safety endpoint</i> excludes deterioration beyond its pre-specified harm margin	supports regression-compatible remodeling; calcium density may rise and is not by itself failure
Stabilization	burden does not materially regress, but vulnerability / cap / necrotic or inflammatory phenotype improves without a new safety signal	supports stabilization, not material regression
Discordant remodeling	one material or biomarker axis improves while a pre-specified critical safety endpoint crosses its harm margin, or another locked load-bearing axis deteriorates materially	not counted as successful coordinated remodeling ; tests the safety-coupling claim directly
Progression	pre-specified burden or vulnerability worsens without a compensating pattern that meets the stabilization rule	negative evidence for the tested configuration
Indeterminate	measurements are insufficient, contradictory beyond the pre-specified rule, or the primary endpoint is missing	cannot be counted as support; requires a new prospectively specified test

14.1 Falsifier F1: Spatial-Confinement Test

The spatial-acidification proposition should be weakened if beneficial mineral resorption can occur without localized proton gradients, or if experimentally confined acidification produces the same cholesterol-efflux impairment and matrix damage as diffuse extracellular acidification.

14.2 Falsifier F2: Coupled-Clearance and Causal-Dependency Test

The coupled-clearance proposition should be weakened if mineral or lipid mobilization improves plaque stability even when the pre-specified co-requisite clearance, efferocytosis, or repair axis is effectively blocked.

In human data, CPRMH does **not** equate coordination with one statistical interaction term. Before outcome access, each stage-specific test must declare the expected causal signature for the relevant edge in the Architecture Lock:

1. **Parallel conditional dependency.** The marginal plaque benefit of improving axis A is attenuated when a pre-specified axis B is severely impaired and amplified when B is permissive, beyond the pre-specified additive model.
2. **Serial causal dependency.** Axis A changes or requires axis B, and the downstream effect of A is materially attenuated when B is experimentally perturbed, functionally disabled, or prospectively shown to carry the causal path under appropriate mediation assumptions [77, 78].

The interaction family or serial-path definition, operational biomarker or latent score, expected direction, covariates, analysis scale, measurement window and adjudication endpoint must be registered before outcome access. A failed interaction cannot be rescued in the same dataset by relabeling the edge as serial mediation, and a failed mediator cannot be replaced by another marker after analysis. If the pre-specified signature repeatedly fails in adequately powered or causally informative

datasets, that dependency must be removed in a prospectively tested revision rather than protected by terminology.

14.3 Falsifier F3: Formation–Resorption Asymmetry Test

If fasting/BHB or mineral inhibitors cause substantial regression of mature hydroxyapatite without measurable changes in resorptive machinery, the proposed asymmetry between easy formation suppression and difficult mature-mineral removal is wrong.

14.4 Falsifier F4: State-Setting Added-Value Test

If fasting, TRE, or exercise provide no explanatory value after controlling for caloric intake, body weight, apoB burden, glycemia, BHB exposure, and circadian timing, they should not remain independent state-setting constructs. **All N7-eligible context modifiers must be locked prospectively under the Context-Lock Rule. A null result within that declared context is negative evidence and cannot be rescued by introducing a post-hoc subgroup or unmeasured “wrong context.”**

14.5 Falsifier F5: Stage-Dependence Test

If early macrophages, advanced human foam cells, and calcified-plaque macrophages respond identically to BHB, GPR109A signaling, efflux stimuli, and pro-resolving cues, the proposed stage-dependence architecture is unnecessarily complex.

14.6 Falsifier F6: Safety-Coupling Test

If increased resorption can occur without any increase in fibrous-cap damage even under high matrix-protease activity, the proposed need to couple clearance with structural repair should be revised.

14.7 Falsifier F7: Relative-Time and Feasibility Test

The sleep-anchored grassroots extension should be weakened if fixed wall-clock timing or direct circadian-phase measures consistently explain metabolic and plaque-relevant outcomes better than sleep-relative timing, or if embedding activity into daily life produces substantially poorer long-term exposure than conventional planned exercise after realistic adherence and time-cost are accounted for. The proposed starch-reduction interval should also be revised if different pre-sleep cutoffs produce no separable physiological effect after total intake, food quality, and energy balance are controlled.

14.8 Falsifier F8: Coordination-Necessity and Functional Single-Axis Sufficiency Test

The central theory should be simplified—and, for the tested disease stage, may collapse—if a single causal axis is **functionally sufficient** for safe regression under the pre-specified adjudication rule. Perfect pharmacological selectivity is not required, because most vascular interventions are pleiotropic. Instead, functional single-axis sufficiency is supported when all of the following are prospectively satisfied:

1. a parsimonious one-axis causal model predicts the safe-regression endpoint;
2. candidate co-requisite axes add no reproducible incremental causal contribution;
3. perturbing, stratifying, or otherwise disabling the presumed co-requisite axes does not materially attenuate the primary-axis effect; and

4. the same sufficiency pattern replicates with an orthogonal intervention or natural experiment that reaches the primary axis through a different upstream route.

If these conditions are met, pleiotropic downstream changes are interpreted as consequences rather than evidence that multi-axis coordination is necessary. If, instead, blocking or disabling a pre-specified co-requisite axis attenuates the primary-axis effect, the corresponding causal dependency remains viable.

14.9 Falsifier F9: Anchor-Stability Test

Among rotating-shift participants, a stable or partially overlapping sleep anchor should outperform full cycle-by-cycle re-anchoring on pre-specified metabolic or vascular state-setting outcomes after matching or adjusting for total sleep duration, energy intake, apoB, and relevant medication exposure. If anchor stability adds no signal, Negative Modality 9 and the Level 0 SAGDRC condition should be removed.

14.10 Falsifier F10: Metabolic–Renal Inclusion Test

If obesity + type 2 diabetes + CKD G3 participants cannot achieve the proposed state-setting exposures without clinically important hypoglycemia, dehydration, kidney-function deterioration, nutritional compromise, or unsustainable burden—or if the adapted branch provides no explanatory value beyond standard risk-factor management—the equity and portability claims of SAGDRC must be narrowed. The biological CPRMH core may survive, but its low-cost translational arm would not be justified for this phenotype.

14.11 Falsifier F11: Architecture-Stability Test

For each plaque stage, the load-bearing set and causal-signature map are locked before outcome analysis. If a pre-specified load-bearing axis or edge repeatedly fails, that result counts as negative evidence for the current architecture. The framework may be revised in a new version, but the failed axis cannot simply be deleted from the same dataset and the remainder declared a success. **If the research programme repeatedly preserves itself only by shrinking or re-labeling the load-bearing architecture after each anomaly, without prospective replication of the revised map, CPRMH should be judged degenerating rather than predictive.**

14.12 Falsifier F12: Negative-Modality Exhaustion Test

For each stage-specific test, the blocker set B_s is locked together with the load-bearing set L_s . The current architecture is falsified for that stage if all of the following are prospectively demonstrated: (1) adequate target engagement of the relevant positive / allied axes, (2) the locked causal signature or signatures occur in the predicted direction, (3) every critical member of B_s is absent, controlled, or below its pre-specified blocking threshold, (4) the structural-safety gate is adequately measured, and (5) the pre-specified stabilization / safe-regression outcome nevertheless fails. An unregistered “unknown blocker” cannot rescue that result in the same dataset. Repetition of this pattern across multiple central plaque stages would weaken the CPRMH core rather than only one translational arm.

14.13 Falsifier F13: Negative-Modality Directionality / Inversion Test

Each negative modality makes a directional claim. If the condition defining a locked negative modality is present with

adequate target engagement, yet its predicted downstream impairment or harm repeatedly fails to appear—or the outcome reproducibly moves in the opposite direction—that negative modality must be weakened, retired, or reclassified in the next prospectively tested version. This rule is especially consequential for N8: if a sustained, large apoB increase were to coexist reproducibly with safe advanced-plaque regression after adequate control of alternative explanations, the hydraulic inflow principle itself would require revision rather than being protected by labeling the phenotype exceptional.

14.14 Falsifier F14: Measurement-Adequacy Lock Test

A failed prediction cannot be rescued by declaring the locked assay, imaging measure, or biomarker inadequate *after* it has passed the pre-specified M_s quality-control and adequacy rules. If independent evidence demonstrates a true measurement failure, the study is measurement-indeterminate and cannot support or refute the affected edge. If no such failure is demonstrated, the observed null, inversion, or exhaustion result remains negative evidence for the architecture. Repeated dependence on post-hoc measurement objections is evidence that the programme is methodologically degenerating.

14.15 Falsifier F15: Threshold-Robustness Test

Target-engagement, blocker, and safety conclusions must be tested across the pre-specified plausible T_s range. If CPRMH is supported only at a narrow cutoff selected because it preserves the predicted architecture, while nearby biologically plausible thresholds reverse the adjudication, the evidence is threshold-fragile and cannot count as strong confirmation. Repeated threshold fragility across independent datasets weakens the corresponding modality or edge.

14.16 Falsifier F16: Causal-Precedence / Non-Circularity Test

A blocker or mediator must be operationally distinct from the downstream outcome it is invoked to explain. If a negative modality can be identified only by observing the very plaque failure that it is supposed to cause, the explanation is circular and inadmissible for causal adjudication. Repeated inability to measure the proposed blocker upstream or independently means that the blocker cannot remain a load-bearing explanatory element of the prospective architecture.

14.17 Falsifier F17: Minimum-Core Collapse Test

The minimum CPRMH core is rejected for a pre-specified established / advanced-plaque domain when **at least two independent or orthogonal causal tests** converge on one of two patterns under adequate M_s , robust T_s , locked safety adjudication, and no post-hoc rescue: **(a)** a functionally sufficient single-axis model produces safe remodeling while candidate co-requisite axes add no necessary causal information; or **(b)** the locked multi-axis architecture achieves target engagement, the relevant blockers are exhausted, and the locked causal signatures are observed, yet the predicted stabilization / safe-regression phenotype still fails and a simpler rival model explains the result better. Under either replicated pattern, the coordination claim for that declared core domain is **falsified**, not merely weakened. A new theory may be constructed, but it is not confirmation of the failed CPRMH core.

15. Boundary Conditions

The CPRMH is explicitly conditional.

15.1 Plaque stage

- early lipid-rich lesions;
- established foam-cell plaques;
- necrotic plaques;
- microcalcified plaques;
- dense fibrocalcific plaques

should not be assumed to share identical clearance capacity.

15.2 Mineral phase

Amorphous or poorly crystalline calcium phosphate may be more labile than mature hydroxyapatite. Mineral-phase composition should therefore be measured rather than inferred from CAC alone.

15.3 Renal function

CKD changes phosphate handling, PPI balance, acid–base physiology, VSMC calcification pressure, inflammation, medication exposure, and mineral metabolism. It is therefore a **modifier of modality availability and dose**, not an exclusion criterion. CKD evidence cannot be generalized automatically to individuals with normal renal function, and evidence from normal renal physiology cannot be assumed to transfer unchanged into CKD. CKD G3 should additionally be stratified by eGFR, albuminuria, laboratory phenotype, nutritional status, and medication context rather than treated as one homogeneous state [71, 72].

15.4 Diabetes, obesity, and metabolic dysfunction

Diabetes and obesity can alter macrophage phenotype, efferocytosis, oxidative stress, lipid handling, glycemic exposure, sleep, physical capacity, and vascular calcification. These changes can make some positive modalities harder to enter while simultaneously increasing the potential value of lowering pathological inflow. The framework therefore includes an explicit metabolic–renal adapted branch rather than assuming that a positive modality observed under normal metabolism is either automatically available or automatically unavailable in diabetes. Weight management, physical activity, and dietary quality remain legitimate allied modalities in people with diabetes, obesity, and CKD when individualized to renal and medication constraints [71, 72, 73].

15.5 ApoB inflow

Clearance cannot be interpreted independently of continued lipoprotein influx. High apoB exposure may overwhelm positive clearance modalities.

15.6 Hepatic clearance

Time-restricted feeding studies demonstrate that hepatic lipoprotein-reuptake capacity can determine whether fasting produces an anti-atherosclerotic effect.

15.7 Shift work, rotating schedules, and circadian misalignment

Sleep-relative timing improves practical portability but does not abolish endogenous circadian biology. Night work can impair glucose tolerance even when total energy intake is unchanged, and experimental redistribution of meals away from the biological night can mitigate some metabolic effects.

Sleep irregularity itself is associated with incident CVD and subclinical atherosclerosis in MESA [74, 75]. Rapidly rotating schedules may therefore remain a negative boundary even when the person follows sleep-relative feeding rules. SAGDRC now treats **anchor stability as a separate exposure**: a stable or partially overlapping sleep anchor is preferred when feasible, while full cycle-by-cycle re-anchoring is retained as a falsifiable comparator rather than assumed to be protective.

15.8 Socioeconomic feasibility and time poverty

A behavior that requires extra travel, equipment, childcare, meal-preparation time, or protected leisure may fail even when biologically effective. Feasibility is therefore part of exposure validity, not an afterthought. The model predicts higher population reach when safe movement is embedded into transport, household activity, caregiving, and other ordinary routines before additional exercise is prescribed.

Occupational physical activity is treated separately from freely chosen movement. A worker who already performs prolonged manual labor may not benefit from adding more work-like activity. Repetitive lifting, static loading, prolonged standing, heat exposure, insufficient recovery, and low control over intensity can create a different cardiovascular and musculoskeletal exposure from self-paced exercise. The model therefore predicts that **movement substitution, pacing, micro-recovery, and added leisure activity where feasible may be more relevant than simply increasing occupational workload**. This is an equity safeguard against converting labor burden into a health prescription.

15.9 Sex, age, genetics, and Lp(a)

Plaque phenotype, calcification burden, and lipoprotein biology vary by sex, age, and genetic background. **ENPP1 and ABCC6** provide causal human anchors for PPI-deficiency-driven ectopic mineralization, while **PCSK9 loss-of-function biology** provides a human natural experiment for lifelong low LDL exposure. **Lp(a)** is a major genetically influenced lipoprotein exposure associated with calcific and atherosclerotic phenotypes and should not be assumed to be corrected by fasting, exercise, or ordinary dietary timing. Sex and age can modify plaque composition, calcification pattern, metabolic response, and feasibility of lifestyle exposures. These variables should therefore be pre-specified effect modifiers, not residual covariates added after analysis.

15.10 Frailty, unintended weight loss, and medical vulnerability

Rapid or unintentional weight loss in older adults can signal frailty or underlying disease and should not be conflated with intentional, gradual metabolic improvement [63]. Fasting protocols also require special caution in people using glucose-lowering drugs that can cause hypoglycemia, during pregnancy, in eating disorders, and in other medically vulnerable states. The grassroots architecture aims to reduce avoidable disease burden, not to replace necessary clinical care.

15.11 Microbiome and bile-acid mediation

Meal timing and time-restricted eating can alter gut microbial composition and circulating or fecal metabolites in humans, but findings are heterogeneous. Recent feeding trials report microbiome/metabolome changes beyond caloric restriction in some settings [60], while other randomized studies report

minimal microbiome change [61]. A 2025 feeding trial also linked timing of unsaturated-fat intake to insulin sensitivity through a gut microbiota–bile-acid axis [62]. CPRMH therefore treats the microbiome as a **possible mediator or rival pathway**, not as a proven mechanism of plaque remodeling.

16. Methodology

16.1 Research design

This article is a **synthesis concept paper** using a **Theoretical Integrative Review (TIR)** architecture [1, 2, 3]. TIR is appropriate because the phenomenon is distributed across multiple bodies of theory and mechanistic research rather than contained within one empirical literature. The objective is not pooled effect estimation, clinical guideline development, or exhaustive systematic review. The objective is to critically examine explanatory components and construct a more coherent theoretical account.

Following conceptual-research guidance, the logic of knowledge creation is made explicit: the paper uses **theory synthesis and conceptual model construction**, not literature aggregation [1].

Search horizon and evidence update. The working corpus was updated through 21 August 2026. Targeted searches were used to locate load-bearing mechanistic anchors, direct contradictory evidence and nearest theoretical neighbors rather than to claim systematic-review saturation. The final update incorporated the August 2026 BMAL1–MERTK fasting-regression study [66], the August 2026 cardiovascular chrononutrition review [67], and the ESC/EAPC scientific statements on coronary plaque regression [68, 69]. Search strings, selection decisions, contradictory items and claim transformations are intended to remain traceable in the living evidence ledger.

Within the TIR, **Perspective Synthesis** is used as the analytic operator.

The synthesis proceeds through five stages:

Locate → Decompose → Friction → Recompose → Stress-test

16.2 Locate

Literature is located across seven mechanistic domains:

1. vascular calcification and calcium-phosphate mineral biology;
2. osteoclast and macrophage mineral resorption;
3. macrophage lipid digestion and cholesterol efflux;
4. efferocytosis and necrotic-core biology;
5. inflammation resolution and fibrous-cap stability;
6. fasting, ketone biology, exercise, and metabolic state transitions;
7. imaging and clinical evidence on plaque regression, calcification density, and stabilization.

The corpus is **mechanism-led and theory-led rather than exhaustive**.

Priority is given to:

- field-defining mechanistic studies;
- human plaque studies;
- causal animal experiments;
- clinical imaging studies;
- recent mechanistic reviews;
- contradictory findings that define boundaries.

16.3 Decompose

Each body of literature is decomposed into:

- unit of analysis;
- plaque material or cell type;
- proposed mechanism;
- spatial scale;
- temporal scale;
- evidence level;
- required exposure/dose;
- disease context;
- measured endpoint;
- interpretation permitted by the endpoint;
- unresolved assumptions.

Every calcification study is additionally classified as addressing:

1. initiation;
2. propagation;
3. maturation;
4. density/phase transformation;
5. persistence;
6. true removal.

This prevents “less formation” from being rewritten as “decalcification.”

16.4 Friction

Every load-bearing proposition must encounter:

- a near-neighbor mechanism;
- a rival explanation;
- contradictory evidence;
- a dose-translation problem where relevant;
- a human–animal translation challenge;
- a disease-stage boundary;
- a measurement limitation.

Contradictory findings are used to specify **boundary conditions**, not treated as noise.

Examples include:

- BHB suppresses vascular calcification, but physiological fasting exposure may be lower than experimental doses;
- fasting reduces atherosclerosis in some LDLR models but not ApoE models;
- local acidification can dissolve mineral, while diffuse acidity impairs cholesterol efflux;
- calcification can increase during statin-mediated plaque stabilization;
- macrophages express CA2 but may lack complete osteoclast-like resorptive capacity.

16.5 Recompose

Only mechanisms with distinct explanatory roles are retained.

The final architecture is not:

BHB + CA2 + ABCA1 + MerTK + RvD1 + statins
but:

**Formation pressure → Local resorption → Lipid export
→ Debris clearance → Resolution → Structural repair**

Every molecular pathway is assigned to one or more functions in this architecture.

16.6 Stress-test

The theory is conceptually stress-tested against conditions including:

- fasting without meaningful ketosis;
- ketosis without caloric restriction;
- weight loss without fasting;
- ApoE versus LDLR deficiency;
- normal renal function versus CKD;
- diabetes versus non-diabetes;
- early microcalcification versus mature hydroxyapatite;
- high apoB influx despite increased efflux capacity;
- improved CAC density despite increased calcium score;
- increased macrophage resorption with simultaneous cap proteolysis;
- full positive-axis target engagement after prospective exhaustion of the relevant negative-modality blocker set;
- inversion of a locked negative-modality prediction, in which the presumed blocker is present but the predicted impairment is absent or reversed.

A theory that cannot specify where it should fail is not sufficiently constrained.

16.7 Evidence discipline

Evidence types are not flattened.

Layer A — physicochemical evidence. Can the material process occur?

Layer B — cell/tissue mechanistic evidence. Can cells execute the proposed process?

Layer C — animal evidence. Can the integrated organism support it?

Layer D — human mechanistic/plaque evidence. Does the machinery exist in human disease?

Layer E — human intervention/imaging evidence. Does modifying the system alter plaque composition or outcome?

Layer F — everyday-exposure evidence. Do fasting, exercise, or diet generate the required physiological exposure in real life?

No lower layer is automatically promoted to a clinical claim.

16.8 Limitations of the synthesis

This is a theory-led integrative synthesis rather than a systematic review or meta-analysis. The corpus is selectively constructed for mechanistic leverage and therefore remains vulnerable to literature-selection bias, single-author interpretation, uneven evidence density across domains, and differential maturity of the component sciences. No claim of saturation is made.

Several parameters are deliberately **hypothesis-generating rather than established guidance**. The proposed S–4 to S–6-hour starch-reduction gate is not a validated clinical threshold. Sleep-relative timing is a pragmatic field anchor rather than a direct measure of endogenous circadian phase, and experimental night-work studies indicate that biological-night eating can remain harmful even when sleep timing is shifted. Evidence among real-world shift workers remains sparse: a 2026 systematic review located only four randomized trials of meal-timing interventions in shift-work settings [41].

The grassroots foods and embedded-activity examples are culturally grounded implementation candidates, not proof that the full CPRMH sequence reduces cardiovascular events. “Embedded activity” also cannot be equated with heavy occupational labor: occupational physical activity may carry different hemodynamic, recovery, and injury profiles from leisure exercise. Finally, the complete chain—relative meal timing → repeated remodeling-permissive states → measurable human plaque stabilization or regression—has not yet been demonstrated in one intervention study. These limitations are not reasons to remove the propositions; they specify where empirical work must be capable of disproving them.

A further limitation is **adjudication and causal-identification dependence**. No universal imaging metric, harm margin, mediator, or interaction scale can classify every experimental system. The manuscript therefore does not pretend that one numerical threshold or one statistical signature applies across IVUS, CCTA, OCT, ex vivo plaque, animal models, and biomarker studies. Instead, each study must lock its target contrast, stage-specific load-bearing set L_s , negative-modality blocker set B_s , measurement model M_s and adequacy criteria, threshold set T_s with plausible sensitivity range, causal-signature map, primary burden endpoint, structural-safety endpoints, harm margins, context modifiers and classification rule prospectively. Causal mediation additionally requires stronger identification assumptions than ordinary association, and statistical noninteraction does not by itself prove absence of biological dependency [77, 78]. Failure to pre-specify these elements re-opens the very flexibility that the falsification architecture is designed to prevent.

16.9 TIR self-audit

To prevent a narrative review from merely adopting the language of a Theoretical Integrative Review, the manuscript explicitly maps its design onto core TIR functions:

16.10 Versioned evidence provenance and living falsifiability

The framework is designed to be maintainable as a **living evidence object** rather than a static list of citations. Each load-bearing proposition should be representable with: claim text, plaque stage, locked load-bearing set L_s , locked negative-modality blocker set B_s , locked measurement model M_s , locked threshold set T_s , blocker operationalization / direction / temporal role, expected causal signature and direction, evidence layer, supporting studies, discordant studies, adjudication endpoint / harm margin, measurement-adequacy status, threshold-robustness status, core-domain status where relevant, unresolved items, and an explicit kill condition. Automated literature retrieval can assist this process but cannot replace human adjudication: conservative keyword-based harm detection can over-flag background sentences, while genuine discordant signals can otherwise be missed. The desired workflow is therefore **machine-assisted retrieval + human reading + versioned adjudication**.

This methodological extension is not evidence for CPRMH itself. It is an epistemic safeguard intended to make a conceptual article continuously falsifiable as the literature changes.

17. Research Questions

RQ1. What determines whether plaque macrophages enter a mineral-resorptive, lipid-exporting, efferocytic, pro-resolving, or maladaptive modality?

RQ2. Can mineral-facing acidification be spatially confined sufficiently to promote demineralization without impairing cholesterol efflux or weakening the fibrous cap?

RQ3. Does suppression of new vascular mineral formation occur more readily than removal of mature hydroxyapatite, and how does this asymmetry vary by mineral phase?

RQ4. Under what conditions do fasting, BHB, exercise, or feeding patterns alter plaque-relevant modalities beyond their effects on calories, weight, glycemia, apoB, and circadian timing?

RQ5. Which combination of positive modalities is necessary for true plaque regression rather than simple stabilization?

RQ6. Should successful remodeling be evaluated through plaque composition, density, inflammation, necrotic core, and cap integrity rather than calcium burden alone?

RQ7. Does a sleep–wake–anchored, low-cost daily cycle of relative meal timing, starch reduction before sleep, sleep–spanning fasting, caffeine withdrawal, embedded movement, and minimally processed local foods generate plaque-relevant metabolic states more reliably across socioeconomic and shift-work contexts than fixed clock-time lifestyle prescriptions?

RQ8. Do human genetic and pharmacologic perturbations that selectively move one modality—PPi deficiency, extreme apoB lowering, SGLT2-related calcification suppression, or cholesterol-efflux manipulation—produce the cross-modality effects predicted by CPRMH, or do they reveal that the proposed coordination architecture is unnecessarily complex?

RQ9. When a primary modality is improved in humans, does severe impairment of a pre-specified second load-bearing modality attenuate the marginal plaque benefit in the predicted direction, and can that dependency classify outcomes more accurately than additive single-axis models under the pre-specified adjudication rule?

RQ10. For each plaque stage, which modalities are truly load-bearing, and does their pre-specified dependency appear as parallel effect modification, serial mediation / blockade, or neither? Can a locked causal-signature map outperform a simpler single-axis model without post-hoc deletion or relabeling of failed axes?

RQ11. Can the negative-modality blocker set for a defined plaque stage be prospectively specified and empirically exhausted? If target engagement and the locked causal signatures are present while every critical blocker is absent or controlled, does failure of safe remodeling falsify the stage-specific architecture as predicted?

RQ12. Do N1–N9 retain their pre-specified directional effects across orthogonal systems, or do some presumed negative modalities repeatedly invert—remaining present while efflux, resolution, structural safety, or regression improve—thereby requiring retirement or reclassification of the negative claim?

Table 11. TIR self-audit

TIR function	How this manuscript implements it
problem distributed across literatures	mineral chemistry, vascular biology, macrophage lipid handling, efferocytosis, resolution, imaging, chrononutrition, physical activity, health equity
integration rather than pooled-effect estimation	no meta-analytic effect claim; mechanisms are reorganized around a shared remodeling problem
declared corpus logic	theory-led, mechanism-led anchors selected for explanatory leverage; no saturation claim
explicit analytic operation	Locate → Decompose → Friction → Recompose → Stress-test
theoretical product	CPRMH + SAGDRC, with positive/negative/allied modalities and a material-specific routing architecture
contradiction handling	contradictory results define boundaries, rivals, and falsifiers rather than being averaged away
translation discipline	evidence Layers A–F prevent lower-level mechanisms from becoming clinical claims
outcome adjudication	pre-specified burden endpoints plus conjunctive safety endpoints and harm margins distinguish safe regression, stabilization, discordant remodeling, progression, and indeterminate results
architecture / causal-signature lock	the stage-specific load-bearing set and expected parallel or serial dependencies are fixed before outcome access; revisions require independent prospective testing
negative-modality lock	the admissible blocker set, operational thresholds and predicted directions are pre-specified; a new blocker cannot rescue the dataset in which it was discovered
measurement / threshold lock	assays, construct validity, QC / adequacy rules and plausible decision-threshold ranges are fixed before outcome access; measurement doubt or cutoff movement cannot rescue a result retrospectively
causal precedence	blocker and mediator constructs must be upstream or analytically distinct from the downstream outcomes they explain, preventing circular negative-modality attribution
minimum core identity	CPRMH retains its name only while at least two separable load-bearing functions show necessary coordination beyond a sufficient single-axis account in the declared core disease domain
reflexivity and limits	single-author interpretation, selective corpus, dose translation, circadian uncertainty, feasibility, and safety are declared

RQ13. Do the target-engagement and blocker conclusions remain unchanged when the same pre-specified biological constructs are measured with orthogonal assays or imaging methods that satisfy the locked M_s adequacy criteria?

RQ14. Are the major CPRMH adjudications stable across the prospectively justified plausible T_s range, or does apparent support depend on a narrow threshold whose selection is not externally defensible?

RQ15. In the declared established / advanced-plaque core domain, can a simpler single-axis or rival causal model repeatedly outperform the locked coordination architecture under adequate measurement, robust thresholds, exhausted blockers and pre-specified safety adjudication? If so, has the minimum CPRMH core been falsified?

18. Empirical Research Program

18.1 Study 1 — Human state-mapping study

Recruit participants undergoing standardized fasting/feeding protocols.

Measure repeatedly: BHB; glucose; insulin; free fatty acids; apoB; triglyceride-rich lipoproteins; phosphate; magnesium; inflammatory mediators; SPM-related lipidomic markers; and monocyte/macrophage transcriptional states where feasible.

Primary aim:

identify whether reproducible human metabolic states corresponding to proposed positive modalities actually occur.

18.2 Study 2 — Serum-transfer VSMC experiment

Expose calcifying human VSMCs to serum collected from the same individuals during the fed state, early fasting, established physiological ketosis, and the post-exercise state.

Determine:

- RUNX2;
- ALP;

- calcium deposition;
- extracellular vesicles;
- apoptosis;
- autophagic flux.

This separates real physiological serum states from pharmacological BHB dosing.

18.3 Study 3 — Human macrophage stage experiment

Compare:

- monocyte-derived macrophages;
- cholesterol-loaded foam cells;
- plaque-derived macrophages where available.

Expose cells to physiological fasting-state conditions.

Measure:

- CA2;
- V-ATPase localization;
- ABCA1/ABCG1;
- MerTK;
- efferocytosis;
- cholesterol efflux;
- lysosomal function;
- SPM response.

This directly tests stage dependence.

18.4 Study 4 — Mineral-interface experiment

Use distinct mineral substrates:

- amorphous calcium phosphate;
- poorly crystalline apatite;
- mature hydroxyapatite;
- calcified extracellular matrix.

Ask whether macrophages form localized acidic resorption zones and whether mineral removal can occur without increased collagen/elastin degradation.

18.5 Study 5 — Coupled-clearance experiment

Construct a model containing:

- mineral;
- aggregated lipoprotein;
- apoptotic cells;
- macrophages;
- VSMC-derived matrix.

Manipulate one clearance modality at a time and in combinations.

The key question is whether **combined positive modalities outperform single-mechanism activation without increasing structural damage**.

18.6 Study 6 — Longitudinal human imaging

A definitive translational study should avoid CAC score as the sole endpoint.

Measure:

- CAC volume;
- CAC density;
- non-calcified plaque;
- low-attenuation plaque;
- fibrous-cap phenotype where imaging permits;
- perivascular inflammation;
- plaque morphology.

This allows distinction among:

progression → stabilization → compositional remodeling → true regression.

18.7 Study 7 — Sleep-anchored grassroots-cycle trial

Compare three pragmatic conditions across conventional day-time workers and shift workers:

1. usual lifestyle;
2. fixed clock-time lifestyle advice;
3. sleep-anchored SAGDRC advice.

The SAGDRC arm would use relative caffeine, starch, last-calorie, and fasting gates plus Embedded Activity First and a locally affordable food matrix. Primary outcomes should include adherence, incremental time cost, incremental financial cost, BHB exposure, glucose/insulin dynamics, apoB, blood pressure, sleep duration/efficiency, and selected inflammatory/resolution markers. The trial should pre-specify strata for conventional schedules, stable night work, rotating shifts, and a metabolic–renal subgroup with obesity + type 2 diabetes + CKD G3. In the latter, the intervention must use the adapted branch rather than identical fasting or protein targets. A longer trial could add CT-based plaque composition outcomes.

A key design requirement is to record **feasibility as an exposure variable**. If a biologically promising regimen cannot be sustained by shift workers or low-resource participants, the theory has failed at the translational level even if efficacy appears under laboratory compliance.

18.8 Study 8 — Stage-locked causal-dependency triangulation

Use existing causal perturbations rather than inventing a single compound expected to perform every task. Candidate contrasts include genetically or pharmacologically altered PPI signaling, intensive apoB lowering, clinically indicated SGLT2-inhibitor exposure, ex vivo cholesterol-acceptor manipulation and targeted perturbation of MERTK/efferocytosis.

Before outcome access, each analysis must register: (1) **plaque stage**, (2) **the load-bearing set L_s** , (3) **the negative-modality blocker set B_s with operational thresholds / directions**, (4) **one primary axis**, (5) **one pre-specified dependency edge**, (6) **whether that edge is expected to appear as parallel conditional dependency, serial dependency, or convergent evidence**, (7) **the direction / mediation path and analysis scale**, and (8) **the burden plus safety adjudication endpoints and harm margins**. The design should preferentially triangulate the same edge with two orthogonal sources of variation, such as pharmacologic exposure plus genetics, or ex vivo perturbation plus longitudinal human imaging.

The primary falsification question is therefore whether the *locked* causal signature appears. A failed interaction does not count as successful mediation in the same dataset unless serial dependency was pre-specified. Conversely, a serial path is strongest when perturbing the proposed mediator materially attenuates the upstream effect. Repeated failure of the locked signature, or functional single-axis sufficiency across orthogonal interventions, argues for a simpler model.

18.9 Study 9 — Metabolic–renal feasibility and interaction study

Recruit adults with obesity, type 2 diabetes, and CKD G3 and compare a renal-adapted SAGDRC sequence with a control condition matched for standard clinical risk-factor management. The design should **not** require prolonged fasting. Pre-register safety stopping rules and medication-management procedures with clinical oversight. Core readouts should include eGFR, albuminuria, electrolytes as clinically relevant, blood pressure, glycemic exposure including hypoglycemia, apoB, body composition / weight trajectory, sleep regularity, achieved fasting duration, activity exposure, and participant time / financial burden. The key theoretical question is whether relative temporal organization adds explanatory or physiological signal after standard CKD and diabetes care is held constant.

18.10 Study 10 — Negative-modality exhaustion challenge

Select one plaque stage for which the necessary measurements are technically credible and lock both L_s and B_s . The preferred first implementation is a multi-component ex vivo / animal or deeply phenotyped human-mechanistic design, not a broad lifestyle trial. Pre-specify target-engagement thresholds for the positive axes, operational thresholds for every critical blocker, the causal signature map, and the outcome / safety adjudication rule. The decisive condition is an **exhaustion cell**: positive target engagement is achieved, every critical member of B_s is absent or controlled, and the locked causal dependencies are observed. If the predicted plaque phenotype still fails to stabilize or regress, the stage-specific architecture is falsified. Any new blocker discovered from that failure is entered only into a subsequent version and must be tested prospectively. A complementary inversion arm deliberately induces or selects a locked negative modality while holding other blockers controlled; failure of the predicted harmful direction tests N1–N9 themselves rather than the positive architecture.

18.11 Study 11 — Measurement-threshold replication and core-collapse challenge

Select a pre-specified established / advanced-plaque domain and freeze the full pre-specification stack (L_s, B_s, M_s, T_s).

Measure at least one central load-bearing construct and one critical blocker with **orthogonal measurement methods** where technically possible (for example, imaging plus ex vivo / biochemical readout, or two assays based on distinct measurement principles). Define adequacy, QC failure and threshold sensitivity before outcome access. The first aim is to estimate whether adjudication is **measurement-stable and threshold-stable**. The second aim is adversarial: compare the locked CPRMH model against the strongest pre-specified simpler rival, including a functionally sufficient single-axis model. A core-collapse claim requires independent or orthogonal replication of the simpler model's advantage or of blocker-exhaustion failure; one anomalous dataset is insufficient to retire the core.

19. Rival Explanations

19.1 Rival 1 — ApoB-only explanation

All apparent remodeling may be explained by reduced atherogenic lipoprotein exposure, with fasting/BHB and other modalities adding little.

Test: hold apoB exposure constant while varying candidate state setters.

19.2 Rival 2 — Weight-loss explanation

Fasting benefits may reflect reduced energy intake and weight rather than metabolic timing or BHB.

Test: isocaloric fasting/TRE versus matched caloric restriction.

19.3 Rival 3 — Glycemic explanation

Improved glucose and insulin may mediate observed changes.

Test: compare interventions that match glycemic improvement but differ in ketosis or meal timing.

19.4 Rival 4 — Stabilization-without-clearance explanation

Clinical benefit may occur through fibrous-cap repair and calcific densification without meaningful mineral or lipid removal.

Test: multimodal imaging plus material-specific biomarkers.

19.5 Rival 5 — Formation-only explanation

All observed calcification benefits may represent lower formation with no resorption.

Test: isotope/mineral tracing or ex vivo quantification of mineral efflux from pre-existing deposits.

19.6 Rival 6 — Circadian-phase explanation

Benefits attributed to sleep-anchored timing may actually be explained by alignment with endogenous circadian phase rather than the sleep-relative architecture itself.

Test: compare sleep-relative timing against dim-light melatonin onset or other circadian-phase markers in controlled shift-work protocols.

19.7 Rival 7 — Adherence/access explanation

A low-cost embedded program may appear superior because people can follow it more consistently, not because its physiological sequence is intrinsically better.

Test: compare biological effects under matched adherence and separately model real-world effectiveness where adherence and time cost are allowed to differ.

19.8 Rival 8 — Microbiome/bile-acid mediation

Meal timing may act primarily through gut-microbial or bile-acid pathways rather than the fasting/BHB or sleep-relative mechanisms emphasized here.

Test: measure microbial and metabolomic mediators and test whether the state-setting effect attenuates when these pathways are modeled [60, 61, 62].

19.9 Rival 9 — Food-quality explanation

The proposed starch-timing gate may add little once total energy, fiber, ultraprocessed-food exposure, saturated fat, sodium and overall dietary quality are matched.

Test: compare sleep-relative starch timing under rigorously matched food quality and energy intake.

19.10 Rival 10 — Open-ended blocker explanation

Any failed remodeling prediction can be made superficially compatible with CPRMH if an unmeasured inhibitor or a new negative modality is allowed to be invented after the result is known. This rival predicts that the apparent explanatory flexibility comes from an expandable blocker vocabulary rather than from true prospective coordination.

Test: freeze B_s before outcome access. A blocker outside the locked set cannot adjudicate the current result; it must be tested prospectively in an independent dataset. If CPRMH repeatedly survives only by adding new blockers after failure, the programme is degenerating rather than predictive.

19.11 Rival 11 — Measurement / threshold rescue explanation

Apparent flexibility could migrate from biology into method: a failed prediction might be preserved by asserting that the assay was insufficient, the proxy missed the true local state, or a different threshold would have classified the blocker or target as present.

Test: lock M_s and T_s , require orthogonal measurement where feasible, report sensitivity across the plausible threshold range, and treat post-hoc assay or cutoff objections as indeterminate hypotheses rather than support for the current dataset.

20. Where the Framework Is Most Likely to Fail

The model should be explicitly weakened if any of the following become well established:

1. **Localized acidification cannot be achieved at vascular mineral interfaces** without diffuse plaque acidification.
2. **Human plaque macrophages lack meaningful mineral-resorptive potential** under all physiologically attainable states.
3. **Advanced foam cells cannot recover cholesterol-export capacity**, making coupled clearance impossible without pharmacological reprogramming.
4. **Efferocytosis cannot keep pace with material mobilization** in advanced plaque.
5. **Everyday fasting/exercise exposures are too small or transient** to alter relevant plaque-cell states.
6. **Observed clinical stabilization is explained fully by apoB reduction** and not by the proposed coordinated modalities.

7. **Dense calcification is consistently harmful rather than conditionally stabilizing** after volume and plaque composition are accounted for.
 8. **Sleep-relative timing adds no explanatory or practical value** beyond endogenous circadian phase and total energy intake.
 9. **The proposed pre-sleep starch-reduction gate has no reproducible physiological effect** after calories, macronutrient composition, and food quality are matched.
 10. **Embedded activity fails to produce meaningful real-world exposure** once occupational fatigue, injury risk, and socioeconomic context are accounted for.
 11. **Context dependence functions only as a post-hoc rescue**: null results repeatedly require newly invented or retrospectively selected N7 contexts that were not pre-specified and do not replicate independently. In that case the state-setting arm is degenerating rather than predictive.
 12. **Human genetics and pharmacologic perturbations do not map onto the proposed modalities**, indicating that the architecture is descriptive rather than mechanistically separable.
 13. **Microbiome or other unmodeled pathways explain the apparent state-setting effects** so completely that the proposed BHB/insulin/relative-time routes add no explanatory leverage.
 14. **Single-modality sufficiency is demonstrated**: an intervention that selectively moves one load-bearing modality, without materially moving the others, repeatedly satisfies the pre-specified *safe regression* adjudication rule in advanced plaque. In that case, coordination is not necessary for that disease stage and CPRMH should collapse toward the sufficient single mechanism.
 15. **Sleep-anchor stability adds no value** beyond total sleep duration, endogenous circadian phase, and total energy intake, including in rotating-shift populations.
 16. **The metabolic–renal adapted branch fails**: obesity + type 2 diabetes + CKD G3 cannot be included without unacceptable safety, nutrition, medication, or feasibility burden, or the adapted branch adds no explanatory leverage beyond standard CKD / diabetes management.
 17. **Architecture drift becomes the only way to preserve the theory**: failed load-bearing axes or dependency signatures are repeatedly deleted or re-labeled after results are known, and revised architectures do not prospectively replicate in independent datasets.
 18. **Safety cannot be demonstrated beyond absence of significance**: critical cap / vulnerability endpoints remain too imprecise to exclude clinically or biologically unacceptable deterioration under pre-specified harm margins. Such results cannot be classified as safe regression.
 19. **Negative-modality drift becomes the rescue layer**: failed predictions are repeatedly explained by N1–N9 or newly proposed blockers that were not included in the pre-specified B_s , and those blockers do not prospectively replicate.
 20. **Negative-modality exhaustion falsifies the architecture**: positive target engagement and locked causal dependencies are demonstrated, every critical member of B_s is absent or controlled, structural safety is adequately measured, yet the predicted stabilization / safe-regression outcome repeatedly fails.
 21. **Negative-modality inversion replicates**: one or more locked negative modalities repeatedly occur without the predicted impairment or with a reproducible opposite-direction outcome, requiring retirement or reclassification of the negative claim.
 22. **Measurement rescue becomes necessary**: failed predictions are repeatedly preserved only by claiming, after outcome access, that assays or imaging readouts that passed their locked adequacy criteria were nevertheless incapable of observing the relevant state.
 23. **Threshold engineering becomes necessary**: the theory survives only at narrow or retrospectively favored blocker / target-engagement cutoffs and fails across the pre-specified plausible biological threshold range.
 24. **Blockers are causally circular**: a negative modality cannot be measured independently or upstream and is identified only by the downstream failure it is invoked to explain.
 25. **The minimum core collapses**: independent orthogonal tests show functional single-axis sufficiency or replicated exhaustion failure in the pre-specified core disease domain, with a simpler rival model outperforming the coordination architecture under adequate measurement and robust thresholds.
- A strong conceptual theory should be able to survive *partial* failure without rewriting the result that produced that failure. If a fasting-related hypothesis or a mineral-resorption branch fails in a locked test, that result remains negative evidence for the tested version. CPRMH may subsequently propose a reduced architecture, but the reduced version is a **new theory version** and must replicate prospectively in independent data before being counted as support. If blocker exhaustion fails across multiple central plaque stages, the core coordination architecture—not merely one branch—must be reconsidered. More decisively, if the pre-specified F17 conditions are met in the declared core disease domain, the minimum coordination core is falsified for that domain and cannot be preserved merely by renaming the revised framework CPRMH.
- ## 21. Claim Architecture: What Is Known, What Is Synthesized, and What Is New
- To protect novelty without overclaiming, CPRMH separates four epistemic levels.
- The absence of direct evidence for the full CPRMH/SAGDRC chain does **not** reduce it to an invalid claim; it identifies it as a hypothesis. The appropriate scientific response is therefore not to collapse the proposition back into already-known lifestyle advice, but to make each novel relation observable, falsifiable, and separable from simpler rivals.
- ## 22. Theoretical Contribution
- The proposed framework makes five conceptual moves.
- ### 22.1 From “plaque cleaning” to material-specific routing
- Mineral, lipid, and dead cells are not one type of waste. Each has a distinct exit route.
- ### 22.2 From acidity to spatially governed acidity
- The biological question is not whether acid can dissolve mineral, but whether proton activity can be confined to the correct interface.

Table 12. Claim architecture: established, synthesized, proposed, and implementation layers

Level	Meaning	Examples in this paper
Established component	supported independently in existing literature	apoB inflow matters; ABCA1/ABCG1 mediate cholesterol efflux; efferocytosis failure expands necrotic core; localized proton secretion can dissolve mineral
Cross-literature synthesis	components are known, but their relation is reorganized around one explanatory object	mineral handling, lipid export, efferocytosis, resolution and cap repair must be considered as coordinated plaque-remodeling tasks
New CPRMH proposition	relationship proposed by this paper and not yet directly demonstrated as a whole	safe remodeling requires a prospectively locked stage-specific architecture: load-bearing set L_s , blocker set B_s , measurement model M_s , threshold set T_s , and parallel / serial causal signatures; the minimum core requires at least two separable functions to add information beyond a sufficient single-axis model
New SAGDRC implementation hypothesis	concrete, falsifiable translation into low-cost daily life	stable-feasible sleep anchoring, relative caffeine / starch / calorie gates, Embedded Activity First, Thai low-cost food substitution, and renal-metabolic adaptation may create recurrent remodeling-permissive windows

22.3 From single mechanism to coordinated modalities

Maximizing one clearance mechanism may create new harm if other modules cannot keep pace.

22.4 From static exposure to state-setting

Fasting, feeding, and exercise are treated as recurrent physiological state changes that may alter the availability of positive or negative modalities.

22.5 From “less calcium” to safer architecture

Plaque composition, mineral density, necrotic-core burden, inflammation, and cap integrity are treated as distinct outcomes.

22.6 From standardized clock time to relative biological time

The practical day is anchored to the person’s main sleep and wake episode, but **anchor stability is now a separate theoretical variable**. This makes the temporal theory testable across conventional schedules, stable night work, and rotating shifts while leaving endogenous circadian phase available as a competing explanation and modifier.

22.7 From wellness consumption to embedded health maintenance

The grassroots extension treats money, additional time, travel, and equipment as intervention costs. It prioritizes the reorganization of ordinary movement, ordinary Thai food, sleep, and non-eating intervals before adding purchasable products or specialized exercise environments.

22.8 From exclusion of high-risk phenotypes to constrained inclusion

Obesity, diabetes, and CKD are not treated as reasons to remove participants from the theory. They are contexts in which modality availability, dose, safety, and measurement change. The model therefore gains credibility only if it can specify **how the intervention architecture adapts without erasing the high-risk population it claims to serve**.

22.9 From measurement flexibility to measurement risk

A mechanism is not prospectively testable merely because a biomarker can be named. CPRMH therefore treats the measurement system itself as part of the falsification architecture: assays, adequacy rules and threshold ranges are declared before outcomes, and a method that passes its own locked criteria cannot be disowned only because the theory loses.

22.10 From expandable framework to minimum core identity

A research programme requires an identity that cannot shrink indefinitely. CPRMH therefore defines an irreducible core: in established / advanced plaque, at least two biologically separable load-bearing functions must exhibit necessary causal coordination beyond a sufficient single-axis explanation. If this proposition fails under replicated orthogonal tests, the core is falsified for that domain.

23. Predictions That Require the Integrated Theory

The framework is only theoretically useful if it generates expectations that do not follow from its component literatures in isolation.

1. **Parallel-dependency prediction.** The benefit of enhancing efferocytosis will be smaller when pathological apoB inflow remains high or when cholesterol export is severely impaired, even if apoptotic-cell uptake itself improves.
2. **Safety-coupling prediction.** Increasing mineral-facing resorption will not constitute favorable remodeling when matrix proteolysis or cap thinning increases in parallel.
3. **Asymmetry prediction.** Interventions can reduce new mineral formation and improve plaque stability without materially removing mature apatite; therefore formation and regression endpoints will frequently diverge.
4. **State-setting interaction prediction.** Fasting or exercise effects will vary by plaque stage, apoB/hepatic-clearance context and circadian alignment, rather than being uniformly beneficial at a fixed duration.
5. **Architecture-over-quantity prediction.** A plaque can become safer while calcium volume remains stable or density rises, provided lipid/necrotic burden falls and cap integrity improves.
6. **Feasibility prediction.** In population translation, an embedded low-cost protocol may generate greater cumulative biological exposure than a more potent laboratory regimen if the latter imposes large time, travel or financial costs; this difference should disappear under experimentally matched adherence.
7. **Human directional effect-modification prediction.** When parallel dependency is the locked signature, the marginal plaque benefit associated with movement of one major axis (e.g., apoB lowering) should be attenuated when a pre-

specified second load-bearing axis is impaired and amplified when it is permissive, beyond additive expectation.

8. **Serial-dependency prediction.** When serial dependency is the locked signature, part of the upstream effect should be carried through the pre-specified downstream axis and materially attenuated when that axis is effectively perturbed; absence of a statistical baseline interaction does not by itself falsify this serial form.
9. **Architecture-lock prediction.** A stage-specific load-bearing map specified before outcome access should replicate more reliably than a sequence of post-hoc architectures produced by deleting failed axes. Repeated need for retrospective shrinkage is evidence against the programme.
10. **Negative-modality exhaustion prediction.** Once the stage-specific blocker set B_s is prospectively exhausted, successful target engagement of the locked positive architecture should still yield the predicted stabilization or safe-regression phenotype; failure at this point directly falsifies the tested architecture.
11. **Negative-modality directionality prediction.** A locked N1–N9 condition should produce its pre-specified adverse / blocking direction; repeated absence or inversion of that direction requires retirement or reclassification of the negative modality rather than contextual rescue.
12. **Measurement-stability prediction.** When M_s is adequate, the qualitative target-engagement / blocker adjudication should replicate across orthogonal measurement approaches; a result should not depend on disowning a locked assay after outcome access.
13. **Threshold-robustness prediction.** The direction of the major architecture adjudication should remain stable across the pre-specified plausible T_s range; support that exists only at a narrow favorable cutoff is threshold-fragile.
14. **Causal-precedence prediction.** Negative modalities should be measurable upstream or independently from the downstream plaque outcome they predict; outcome-defined blockers should add no admissible causal support.
15. **Core-collapse prediction.** If orthogonal prospective tests in the declared advanced-plaque core domain repeatedly favor functional single-axis sufficiency or replicated exhaustion failure under adequate measurement and robust thresholds, the coordination core should be rejected rather than repeatedly revised.
16. **Safety-margin prediction.** Interventions classified as safe regression should exclude unacceptable deterioration on every pre-specified critical vulnerability endpoint using confidence bounds and harm margins; mere non-significance of harm is insufficient.
17. **Anchor-stability prediction.** Under rotating work, preserved sleep-anchor overlap should predict more favorable state-setting exposure than full re-anchoring to each shift after total sleep, calories and apoB are accounted for.
18. **Metabolic–renal inclusion prediction.** The direction of beneficial state-setting should remain observable in obesity + type 2 diabetes + CKD G3 after renal-specific dose adaptation, even though achievable fasting duration, diet composition and modality magnitudes may differ from metabolically healthy participants.

These predictions create the empirical burden of proof for CPRMH. They must be judged under the Context-Lock

Rule and the full pre-specification stack (L_s, B_s, M_s, T_s), together with the locked causal-signature map and outcome-adjudication hierarchy. If the locked dependencies do not outperform simpler additive or single-axis explanations, if blocker exhaustion fails despite adequate measurement, if conclusions are threshold-fragile, or if safe-regression claims cannot pass their safety harm margins, the framework should be compressed or, where F17 is met, its minimum core rejected rather than protected by terminology.

24. From Mechanism to a Low-Cost Health-Maintenance Architecture

The practical implication of CPRMH is not that individuals should attempt to chemically dissolve plaque through food or fasting. It is that ordinary life can be studied as a **recurrent vascular state-setting environment**.

The grassroots extension therefore applies four design principles.

24.1 Embedded Activity First

Movement should first be recovered from activities people already perform. A person who walks to transport, cleans a home, carries goods, works standing, climbs stairs, or moves through caregiving is not “doing nothing” simply because that movement is not branded as exercise. Planned exercise remains valuable, but it is a second layer.

This matters because the opportunity to exercise is socially distributed. Time poverty, shift work, care obligations, unsafe environments, and transportation can make a 30-minute dedicated exercise session materially more expensive for one person than another; this extends the author’s earlier opportunity-structure account of unequal sporting possibility into preventive-health design [45]. A low-cost cardiovascular framework should therefore count **incremental time cost** alongside biological efficacy.

24.2 Relative Timing Instead of Middle-Class Clock Time

A universal instruction such as “stop eating at 18:00” or “stop starch at 14:00” silently assumes a particular workday. SAGDRC instead asks four personal questions:

- When is your principal sleep episode?
- When do you usually wake from it?
- How many hours before sleep can caffeine reasonably stop?
- How far before sleep can starch and then calories be progressively withdrawn without making the routine socially or nutritionally unsustainable?

This allows the same theory to be used by a daytime office worker, market vendor, factory worker, nurse, driver, caregiver, or night-shift worker.

24.3 Ordinary Thai Food Before Health Products

The food architecture is deliberately compatible with low-cost local diets. Its central resources are vegetables, legumes, tofu, eggs, local fish, ordinary rice with less-refined grain substitution where feasible, seasonal fruit, and water. Its primary removals are repeated sugar-sweetened beverages, heavily processed snacks and meats, excessive sodium, and repeated starch-heavy eating near the main sleep episode.

This approach is aligned with contemporary cardiovascular dietary guidance [35, 43, 44] but adds a different theoretical

question: **how should the same inexpensive foods be temporally arranged around wake, activity, and sleep to alter recurrent remodeling modalities?**

24.4 Reduce Dependence by Reducing Preventable Load, Not by Rejecting Care

The public-health ambition is to reduce the amount of avoidable cardiometabolic disease that reaches expensive clinical stages. It is not to imply that plaque disease, diabetes, hypertension, acute coronary syndromes, or high-risk lipid disorders should be self-treated without medical evaluation.

The strongest equity claim is therefore structural:

If the theory is correct, some proportion of cardiovascular risk modification may be achievable by reorganizing time, movement, eating sequence, sleep, and ordinary local foods—resources already present in daily life—rather than by making prevention depend first on commercial products, specialized facilities, or high-cost programs.

This claim is empirically testable through adherence, cost, biomarker, and eventually imaging outcomes.

24.5 A one-cycle example using relative time

A person wakes at *W*. Early in the waking period, substantial meals including starch can occur, preferably with vegetables and protein. Movement is accumulated through transport, work, household tasks, and short post-meal activity. Around *S–8 to S–9 hours*, caffeine ends. Around the proposed *S–4 to S–6-hour* range, starch-dominant foods are reduced. Around *S–2 to S–3 hours*, caloric intake ends when feasible. Fasting continues through the main sleep episode and may extend briefly after waking before refeeding.

The exact wall-clock times change with the person's life. The **sequence does not**.

25. Conclusion

Atherosclerotic plaque remodeling is unlikely to be safely achieved by a single “cleaning” mechanism. The literature instead supports a distributed architecture in which different plaque materials require different handling pathways and in which the safety of clearance depends on coordination.

The strongest current evidence supports several separate propositions: vascular mineral formation can be biologically suppressed; macrophages can generate localized acidic extracellular compartments and can demineralize ectopic calcium under some conditions; macrophages export cholesterol through molecular efflux pathways; defective efferocytosis promotes necrotic-core expansion; pro-resolving mediators can improve efferocytosis and fibrous-cap phenotype; and clinical plaque stabilization can occur while calcification density increases.

These findings also impose strong constraints. Diffuse plaque acidity is potentially pro-atherogenic. Human plaque macrophages may have incomplete osteoclast-like resorptive programs. Advanced foam cells can lose cholesterol-export responsiveness. Mature hydroxyapatite can persist. Increased resorptive protease activity could threaten the fibrous cap. Fasting and BHB effects are context dependent and cannot be equated with direct plaque clearance.

The **Coordinated Plaque Remodeling Modality Hypothesis** therefore proposes that the relevant therapeutic or physiological state is not one that maximizes acidity, ketosis, mineral

dissolution, or macrophage activity. It is one that shifts the system toward **less pathological inflow and new deposition, molecular lipid export, efficient cellular-debris clearance, active resolution, spatially confined mineral handling where feasible, and structural repair that keeps pace with removal.**

Within this architecture, fasting, movement, food composition, stimulant timing, and sleep become scientifically interesting not as detergents but as **state-setting exposures**. The sleep-anchored grassroots extension adds a further proposition: these exposures may be organized by relative biological sequence rather than universal wall-clock time, allowing the same framework to be tested in conventional workers and shift workers while explicitly measuring affordability, time cost, and adherence.

The framework is therefore intended to generate an empirical program rather than a lifestyle prescription. Its success should be judged by whether stage-specific load-bearing architectures and their pre-specified causal signatures replicate prospectively, whether the corresponding negative-modality blocker sets can be prospectively measured and exhausted without post-hoc rescue, whether the locked measurement model and threshold set remain adequate and robust, whether simpler single-axis models fail where CPRMH predicts dependency, whether safety claims exclude unacceptable structural harm rather than merely fail to detect it, and whether the framework explains why some interventions produce progression, some stabilization, some discordant remodeling, and only a subset safe regression. Crucially, CPRMH now carries a minimum identity condition: if adequately measured, threshold-robust and independently replicated tests in the declared advanced-plaque core domain show that a single causal axis is sufficient or that the locked multi-axis architecture fails despite exhausted blockers, the coordination core is falsified for that domain rather than protected through a new blocker, assay objection, threshold shift or retrospective architectural revision.

Declarations

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Draft-reference note. This list prioritizes the load-bearing studies used to construct the conceptual architecture. Bibliographic formatting should be standardized to the target journal before submission.

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Appendix A. Claim-Discipline Rules for Future Revisions

1. **“Reduces calcification” must specify the endpoint.**
Calcium deposition in cells, arterial calcium content, CAC progression, calcium density, and established mineral regression are not interchangeable.
2. **“Fasting effect” must specify achieved physiology.**
Record BHB, energy intake, apoB, glycemia, weight change, and timing.
3. **“Acid-mediated clearance” must specify spatial scale.**
Systemic acidification, diffuse extracellular plaque acidity, and sealed mineral-facing acidification are distinct states.
4. **“Macrophage clearance” must specify material.**
Mineral resorption, lipid digestion, cholesterol efflux, and efferocytosis are distinct functions.
5. **“Improved plaque” must specify architecture.**
Include at least plaque volume, lipid/necrotic burden, calcific volume/density, inflammatory state, and cap stability where possible.
6. **Preclinical evidence must remain preclinical.**
No cell or animal mechanism is to be rewritten as a dietary or fasting recommendation without direct human evidence.

7. Relative-time parameters must be labelled by status.

The caffeine cutoff is evidence-informed; the 4–6-hour starch-reduction gate is a CPRMH/SAGDRC hypothesis parameter; the 2–3-hour last-calorie gate is a pragmatic starting rule. These must not be presented as equally established facts.

8. Feasibility is part of causal translation.

A regimen that works only when participants possess abundant free time, money, transport, or equipment cannot be generalized as a low-cost population strategy without measuring those resources.

9. Shift work requires three-level interpretation.

Sleep-relative timing is a practical anchor, anchor stability is a distinct exposure, and endogenous circadian phase remains a biological modifier and rival explanation.

10. CKD / diabetes adaptation must not become exclusion.

Obesity, type 2 diabetes, and CKD G3 remain inside the translational model, but fasting duration, protein, sodium / potassium / phosphorus decisions, medication safety, hydration, and monitoring must be individualized rather than copied from the metabolically healthy branch.

11. A negative modality must be prospectively admissible.

For a given stage / contrast, specify the relevant B_s , operational marker, threshold or decision rule, expected direction, temporal window and role before outcome access. An unspecified N1–N9 cannot be added later to reinterpret a null or harmful result as support.

12. A new blocker belongs to the next theory version.

Discovery of an unmodeled blocker is scientifically valuable but hypothesis-generating. It cannot adjudicate the dataset in which it was discovered; prospective replication is required before the revised blocker set can support CPRMH.

13. Measurement adequacy must be locked before outcome access.

Name the construct, assay / imaging method, QC limits, repeatability or precision rule, spatial / temporal scale, missingness rule and adequacy floor. A method that passes these rules cannot be disowned after the theory loses without independent evidence of assay failure.

14. Decision thresholds must survive a plausible-range sensitivity test.

Target-engagement, blocker and harm thresholds require a pre-specified rationale and plausible range. A conclusion that survives only at a narrow data-fitted cutoff is threshold-fragile rather than confirmatory.

15. A blocker cannot be its own outcome.

Negative modalities and mediators must be measured upstream, earlier, or through constructs distinct from the downstream endpoint they are invoked to explain.

16. The CPRMH core has a stopping rule.

In the declared established / advanced-plaque domain, replicated orthogonal evidence for functional single-axis sufficiency or exhaustion failure under adequate measurement and robust thresholds falsifies the minimum coordination core; it may motivate a new theory but cannot be counted as confirmation of CPRMH.

Appendix B. Minimal Causal Readout for a Future Human Study

A human study claiming support for coordinated plaque remodeling should ideally measure at least one marker from each layer:

The model should be considered supported only if state-setting exposure is linked to downstream modality changes, not merely to a favorable final imaging association.

Table 13. Minimal causal readout for a future human study

Layer	Candidate readout
State-setting	fasting duration, BHB AUC, insulin/glucose, apoB
Mineral formation	calcification propensity / VSMC-related markers
Mineral structure	CAC volume + density, PET/CT microcalcification where available
Lipid export	HDL cholesterol-efflux capacity / macrophage efflux proxy
Efferocytosis/resolution	SPM profile, soluble MerTK-related markers where validated
Inflammation	hsCRP plus pathway-specific markers
Structural plaque	CCTA low-attenuation plaque, fibrous components, IVUS/OCT subset
Clinical safety	ischemic events, thrombosis, arrhythmia, renal/metabolic adverse events
Measurement adequacy	assay / imaging QC, precision / repeatability, missingness, spatial / temporal resolution and pre-specified adequacy classification for each load-bearing or blocker construct
Threshold robustness	target / blocker / harm threshold rationale plus sensitivity across the prospectively plausible range

Appendix C. TIR Compliance Checklist

Table 14. TIR compliance checklist

Requirement for a defensible integrative conceptual review	CPRMH implementation
Explicit problem	Safe plaque remodeling requires coordination across material-specific processes
Reason integration is needed	Relevant mechanisms are distributed across mineral, lipid, immune, structural, metabolic, temporal, and equity literatures
Declared review type	Theoretical Integrative Review / synthesis concept paper; not systematic review or meta-analysis
Declared corpus logic	Theory-led and mechanism-led; no saturation claim
Analytic procedure	Locate → Decompose → Friction → Recompose → Stress-test
Contradictory evidence retained	Fasting model dependence, diffuse-acid harm, foam-cell failure, ketogenic apoB response, stable calcification, shift-work circadian conflicts
New theoretical product	CPRMH + SAGDRC
Falsifiability	F1–F17 plus explicit architecture, blocker-exhaustion, measurement / threshold, causal-precedence, core-collapse and single-modality sufficiency kill conditions
Translation boundary	Evidence Layers A–F and explicit distinction between hypothesis parameters and established guidance
Reflexivity	Single-author and selection limitations declared; AI is epistemic support rather than theoretical authorship

Appendix D. Evidence-Status Matrix for the Standalone Theory

Table 15. Evidence-status matrix for the standalone theory

Claim	Current status
Acid promotes calcium-phosphate dissolution	established physicochemical principle
Osteoclasts use sealed acidic resorption lacunae	established cellular biology
Plaque/ectopic macrophages can express CA2 and demineralize ectopic mineral	mechanistic / preclinical support
Human plaque macrophages execute complete osteoclast-like resorption	uncertain / likely incomplete
Diffuse acidic plaque environment can impair cholesterol efflux	human-cell mechanistic support
ABCA1/ABCG1 export is a plausible safe lipid-removal route	established macrophage biology
Efferocytosis failure enlarges necrotic core	strong mechanistic/animal + human-plaque support; fasting-linked BMAL1–MERTK activation now has baseline-controlled mouse regression evidence
SPM pathways promote resolution and cap-stabilizing phenotypes	human association + experimental causal support
Intensive apoB lowering can regress human coronary atheroma	randomized human imaging support
Less total calcium is always better	rejected by current imaging/pathology evidence
BHB suppresses vascular calcification	multiple preclinical mechanisms + observational human association
Ordinary fasting removes mature human plaque mineral	unproven
Sleep-relative fasting is superior to clock-time or circadian-phase timing	new CPRMH/SAGDRC hypothesis
S – 4 to S – 6 h starch-reduction gate	new hypothesis parameter
S – 8 to S – 9 h caffeine cutoff	evidence-informed sleep rule, not plaque-specific evidence
Embedded Activity First improves population reach	strong feasibility logic + WHO-compatible activity definition; must exclude occupational-overload interpretations; CPRMH-specific vascular effect unproven
Thai low-cost feeding matrix directly regresses plaque	unproven; proposed implementation of established dietary-pattern principles
Sleep-anchor stability improves cardiovascular or plaque outcomes	observational sleep-regularity support; SAGDRC-specific intervention hypothesis remains unproven
Obesity + type 2 diabetes + CKD G3 can be included in SAGDRC	conceptually supported by current CKD / diabetes lifestyle guidance only through a renal–metabolic adapted branch; plaque-specific benefit remains unproven
Stage-specific negative-modality blocker sets can be prospectively exhausted	new CPRMH methodological / causal hypothesis; required for preventing negative modalities from functioning as post-hoc rescue explanations
Locked measurement and threshold systems can prevent methodological rescue	new CPRMH falsification-discipline proposition; requires prospective validation and orthogonal replication
Minimum CPRMH core can be rejected by replicated single-axis sufficiency or exhaustion failure	explicit new core-collapse criterion; not yet empirically tested as a whole
Full SAGDRC cycle causes plaque regression	central new hypothesis; requires testing; no human plaque-regression trial yet supports the complete cycle

Appendix E. Sleep-Relative Implementation Examples

These examples illustrate the **relative-time rules**. They are not universal prescriptions.

Example 1 — Conventional daytime worker. If principal sleep begins at 22:30:

- caffeine gate: approximately 13:30–14:30;
- starch-reduction gate: approximately 16:30–18:30;
- last-calorie gate: approximately 19:30–20:30;
- fasting spans the sleep episode and may extend after waking according to feasibility and safety;
- embedded movement is accumulated during transport, work, household activity, and short post-meal movement;
- planned aerobic or strength activity is added only where feasible.

Example 2 — Permanent night worker. If principal sleep begins at 09:00:

- caffeine gate: approximately 00:00–01:00;

- starch-reduction gate: approximately 03:00–05:00;
- last-calorie gate: approximately 06:00–07:00;
- fasting spans daytime sleep;
- large meals during the biological night are additionally treated as a circadian risk even when sleep-relative gates are followed.

Example 3 — Time-poor caregiver or informal worker. The first intervention layer is not a gym session. It is to:

- use walking already required for transport or errands;
- distribute housework and carrying tasks to break long sitting where feasible;
- attach short walking to meals or phone calls;
- preserve the caffeine and pre-sleep feeding gates;
- use affordable local foods rather than adding supplements or specialty foods;
- add structured exercise only when time and recovery capacity allow.

The theory predicts that this design may achieve **greater repeated exposure at lower incremental cost**, which must be measured rather than assumed.

Example 4 — Obesity + type 2 diabetes + CKD G3. This example illustrates **adaptation**, not a treatment prescription. The individual keeps the same conceptual sequence—stable-feasible sleep anchor, reduced late substrate loading, sleep-spanning low-input interval, movement, recovery—but prolonged fasting is not mandatory. Food structure is adjusted to renal nutrition requirements rather than converted into a high-protein or ketogenic pattern. Potassium and phosphorus decisions are individualized rather than universally restricted. Research monitoring explicitly includes kidney function, albuminuria, blood pressure, glucose / hypoglycemia exposure, medication class, nutritional status, and feasibility. The branch fails if achieving the temporal sequence requires unsafe medication changes, dehydration, nutritional compromise, or an unsustainable time / financial burden.

Appendix F. Proposed Living Evidence Ledger

Each proposition should be stored as a record containing: proposition ID; exact claim; plaque stage; locked load-bearing set L_s ; locked negative-modality blocker set B_s ; locked measurement model M_s (construct, assay / imaging method, QC, precision, spatial / temporal resolution, adequacy rule); locked threshold set T_s (target-engagement, blocker and harm thresholds plus plausible sensitivity range); blocker direction / temporal role and causal-precedence statement; causal-signature family; evidence layer (A–F); supporting PMIDs/DOIs; discordant PMIDs/DOIs; population/model; exposure/dose; measured endpoint; pre-specified adjudication class; interpretation allowed; unresolved uncertainty; falsifier/kill condition; core-domain status where relevant; last human adjudication date; and version history.

Automated retrieval should be deliberately conservative. A machine flag is not equivalent to a harmful finding; it is an item requiring human reading. Conversely, human adjudication should never silently delete genuine discordance. The ledger therefore operationalizes the article’s epistemic rule: **contradiction defines boundary conditions until evidence justifies stronger revision.**